V.F.S. v2.0 (Open-Gate)

Folding, Stability, and Structure Formation

A Companion Volume

Derived companion to the foundational layer
(core · Lyapunov · geometry · cosmology · theological · Ricci)

Contents

Notation and Conventions

This volume collects the folding, stability, and structure-formation results derived as a companion to the foundational V.F.S. layer. Throughout, *Propositions* are closed-form results; *Corollaries* are interpretive readings in the symbolic register, not independent metaphysical claims; *Remarks* and italicised notes are commentary. All stability results are **domain-conditional**: they hold on the positively-invariant active domain and assert set stability with partial convergence (no global or automatic stability, and no equilibrium — epektasis drives $\Omega_P \to \infty$).

Core symbols.

Ω_P	Brim / scale factor; $\dot{\Omega}_P = \alpha\lambda$, Hubble $H = \alpha\lambda/\Omega_P$
$\lambda,\ \sigma$	Sophia (transformed resistance + grace); resistance
$u = \sqrt{VF + \varepsilon}$	synergy of Voluntas V and Factum F; threshold $\Lambda_c = \gamma/\delta$
$I_{\text{gate}},\ \mathcal{G}_{\text{recepta}}$	Open-Gate grace inflow; received-grace budget
q_{fold}	folding (shape) amplitude; order parameter of the $\mathbb{Z}_2$ morphology
$A_{\text{fold}},\ B_{\text{fold}}$	folding activation index and quartic coefficient; $Q_{\text{fold}} = A_{\text{fold}}/B_{\text{fold}}$, $q_* = \sqrt{Q_{\text{fold}}}$
$\mathcal{L}_{\text{VFS}},\ \mathcal{L}_{\text{ext}}$	Lyapunov certificate; folding-extended certificate
$h,\ \mathbb{H}^2$	comoving hyperbolic spatial metric, $K_h = -1$, curvature radius R_c
$\mathcal{I}(t) = \int_0^t dt'/\Omega_P^2$	expansion / activity budget; finite $\mathcal{I}_\infty < \infty$
$D_q,\ \ell_b$	folding diffusivity (bending stiffness); bending length $\ell_b = \sqrt{D_q/a_0}$
$\Theta,\ Q_{\text{crit}}$	folding-response coefficient $c_0 - c_1(\gamma + k_\sigma)$; reset critical level
λ_{form}	Sophia embodied in form; balance $\sigma + \lambda + \lambda_{\text{form}} = C_0 + \mathcal{G}_{\text{recepta}}$
η_{fold}	folding feedback coefficient (distinct from any core-system η)
Δ_h	Laplace–Beltrami on $\mathbb{H}^2$; live-domain $\Delta_g = \Omega_P^{-2}\Delta_h$

Two activation-index conventions. The activation index appears in two forms across the layers, reconciled in the Folding–Lyapunov Addendum:

$$\underbrace{A_{\text{fold}} = c_0\lambda + c_1\dot{\lambda}^{(0)} - a_0, \quad \dot{\lambda}^{(0)} = -\dot{\sigma} + I_{\text{gate}}}_{\text{canonical (core rate)}} \qquad \underbrace{A_{\text{fold}} = (c_0 - c_1\gamma)\lambda + c_1 k_\sigma \sigma + c_1 I_{\text{gate}} - a_0}_{\text{geometric (minimal rate)}}.$$

The canonical (core-consistent) form is primary; the geometric parameters (k_σ, γ) are an effective linear surrogate. The closed forms $\Theta,\ Q_{\text{crit}}$ are stated in the geometric convention; the invariance and reset-bound results are convention-independent.

The folding diffusivity is structural, $D_q = a_0\ell_b^2$ (rigidity $\times$ bending length^2), and the independent Sophia diffusivity is zero, $D = 0$.

The Foundational Layer (Summary)

This volume sits atop six foundational documents, summarised here so it reads self-contained.

Core. V.F.S. (Volo. Facio. Sum.) is a hybrid dynamical system on the live domain in the coordinates Voluntas V, Factum F, Sophia λ, with synergy $u = \sqrt{VF + \varepsilon}$, resistance σ, Pleroma P and Brim Ω_P. The Open-Gate law $\dot{\lambda} = -\dot{\sigma} + I_{\text{gate}}$ admits a bounded grace inflow; the closed Microcosm is the limit $I_{\text{gate}} \to 0$. The Filtrum Lucis softplus Φ yields the Paschal triad ($\Phi'''(0) = 0$ midway between symmetric zeros of $\Phi^{(4)}$). Resurrectio is a hybrid re-entry map.

Lyapunov. On the active domain $\mathcal{D}_A$ the certificate obeys $\dot{\mathcal{L}}_{\text{VFS}} \le C - C_1\mathcal{L}_{\text{VFS}}$ with $C_1 = \min\{\kappa_\sigma, \kappa_\Delta\} > 0$ from two sign-definite contracting modes (cleansing, will–action alignment). Grace enters C, never C_1. The result is set stability with partial convergence ($\sigma, D_\Delta \to 0$), forward completeness, and non-Zeno resets — domain-conditional, not global.

Geometry. The live domain is hyperbolic: with the canonical warp the scalar curvature is constant and negative, $R = -2/\Omega_P^2$, curvature radius Ω_P. The Brim flow is a forced (not intrinsic) Ricci-like conformal flow, resonant only on $\alpha\lambda\Omega_P = 1$. Sophianic Folding is a supercritical pitchfork of the shape operator producing the amplitude q_{fold} and curvature pockets $R = -2/\Omega_P^2 + \rho_1 Q_{\text{fold}}$.

Cosmology. The Lorentzian lift is a $2{+}1$ open FLRW spacetime, $ds^2 = -dt^2 + \Omega_P^2 h$, with Ω_P as scale factor. Reconstructed content gives $\rho_{\text{eff}}, p_{\text{eff}}$ and an equation-of-state history; Sophia is modal (level / rate / embodiment). In $2{+}1$ the strong energy condition reduces to $p_{\text{eff}} \ge 0$.

Theological & Ricci. The theological layer reads the mathematics in the symbolic register with explicit domain-conditional disclaimers; the Ricci layer disciplines the Perelman analogy (Brim-flow comparison, surgery $\leftrightarrow$ Resurrectio), labelling each correspondence by its strictness.

Part I

Folding and Stability

Chapter 1

What Triggers Sophianic Folding

Scope and epistemic status

This note isolates a single question that the geometric, cosmological, and folding material left implicit: *what is the primary cause of Sophianic Folding?* Three candidate answers are tested — continuous monotone growth of Sophia, the discrete Resurrectio push, and an intrinsic instability of the surface-form itself. The conclusion is that the third is the correct *level* of description, and the first two are interchangeable *channels* to it. As in the parent files, every *Proposition* is a closed-form result; every *Corollary* is an interpretive reading in the symbolic register, not an independent metaphysical claim. Numerical values, where used, are illustrative: the genuine folding parameters $(a_0, b, c_0, c_1, k_\sigma, \eta_{\text{fold}}, \rho_1, q_{\max})$ are not fixed by the source files.

The objects are imported from the integrated geometric/cosmological layer. The shape-mode amplitude q obeys the Landau gradient flow $\dot{q} = A_{\text{VFS}} q - B_{\text{VFS}} q^3$, with

$$\boxed{A_{\text{VFS}} = (c_0 - c_1\gamma)\lambda + c_1 k_\sigma \sigma + c_1 I_{\text{gate}} - a_0, \qquad B_{\text{VFS}} = b + 2c_1\eta_{\text{fold}}\lambda, \qquad Q_{\text{fold}} = \frac{A_{\text{VFS}}}{B_{\text{VFS}}}, \quad q_*^2 = Q_{\text{fold}}.} \tag{1.1}$$

The Sophia-production law that closes the loop is $\dot{\lambda} = k_\sigma \sigma + I_{\text{gate}} - \gamma\lambda - \eta_{\text{fold}} q^2 \lambda$, the Brim law is $\dot{\Omega}_P = \alpha\lambda$, and the Resurrectio reset (core/Lyapunov) is

$$\sigma^+ = q_R \sigma^-, \qquad \lambda^+ = \lambda^- + (1 - q_R)\sigma^-, \qquad \Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}^-_{\text{recepta}}, \qquad (V, F) \text{ continuous,} \tag{1.2}$$

with $0 \leq q_R < 1$, $\kappa_R > 0$, and $u^+ = u^-$.

1.1 The single trigger: a shape-operator bifurcation

The folded amplitude is the order parameter of the one-mode shape potential

$$\mathcal{E}_1(q) = -\tfrac{1}{2} A_{\text{VFS}} q^2 + \tfrac{1}{4} B_{\text{VFS}} q^2 \cdot q^2 = -\tfrac{1}{2} A_{\text{VFS}} q^2 + \tfrac{1}{4} B_{\text{VFS}} q^4, \qquad B_{\text{VFS}} > 0. \tag{1.3}$$

Writing the first shape-stability eigenvalue at the old form $q = 0$,

$$\mu_1^0 := \mu_1\big|_{q=0} = a_0 - c_0\lambda - c_1\dot{\lambda}^0, \qquad \dot{\lambda}^0 = k_\sigma\sigma + I_{\text{gate}} - \gamma\lambda, \tag{1.4}$$

one has the identity $A_{\mathrm{VFS}} = -\mu_1^0$ (the embodiment feedback $-\eta_{\mathrm{fold}} q^2 \lambda$ vanishes at $q = 0$ and is absorbed into B_{VFS}). The control of the whole morphodynamics is therefore the scalar A_{VFS}.

Proposition 1.1 (Folding is a supercritical pitchfork)**.** *For $B_{\mathrm{VFS}} > 0$ the gradient flow $\dot{q} = A_{\mathrm{VFS}} q - B_{\mathrm{VFS}} q^3$ has the old form $q = 0$ as its only equilibrium, locally stable, when $A_{\mathrm{VFS}} < 0$. At $A_{\mathrm{VFS}} = 0$ it undergoes a supercritical pitchfork bifurcation: for $A_{\mathrm{VFS}} > 0$ the old form becomes unstable and two stable folded branches appear,*

$$\boxed{q = 0 \text{ stable} \iff A_{\mathrm{VFS}} < 0, \qquad q_\pm = \pm\sqrt{A_{\mathrm{VFS}}/B_{\mathrm{VFS}}} \text{ stable} \iff A_{\mathrm{VFS}} > 0.} \tag{1.5}$$

Proof. $f(q) = A_{\mathrm{VFS}} q - B_{\mathrm{VFS}} q^3$, $f'(q) = A_{\mathrm{VFS}} - 3B_{\mathrm{VFS}} q^2$. At $q = 0$, $f'(0) = A_{\mathrm{VFS}}$, so $q = 0$ is stable iff $A_{\mathrm{VFS}} < 0$. For $A_{\mathrm{VFS}} > 0$ the nonzero equilibria solve $q_*^2 = A_{\mathrm{VFS}}/B_{\mathrm{VFS}}$, where $f'(q_*) = A_{\mathrm{VFS}} - 3A_{\mathrm{VFS}} = -2A_{\mathrm{VFS}} < 0$: stable. The crossing at $A_{\mathrm{VFS}} = 0$ is the pitchfork. □

Corollary 1.1 (The trigger is loss of stability, not a "push")**.** *The primary cause of folding is the sign change of A_{VFS} (equivalently μ_1^0 crossing zero from above): the old symmetric form ceases to be a potential minimum and becomes a maximum. Past the threshold folding is dynamically inevitable, since $q = 0$ is a repeller; before it, no shape perturbation grows. Folding is therefore an intrinsic property of the surface's stability spectrum, tuned by the state variables, not an external impulse.*

The fundamental object is the bifurcation $A_{\mathrm{VFS}} = 0$. Everything else — continuous growth, the Resurrectio jump — is a means of moving A_{VFS} across that single threshold.

1.2 Decomposition of the driver

The activation index (1.1) is a sum of four physically distinct contributions:

$$\boxed{A_{\mathrm{VFS}} = \underbrace{(c_0 - c_1\gamma)\lambda}_{\text{accumulated Sophia}} + \underbrace{c_1 k_\sigma \sigma}_{\text{kathartic production}} + \underbrace{c_1 I_{\mathrm{gate}}}_{\text{received grace}} - \underbrace{a_0}_{\text{old-form rigidity}}.} \tag{1.6}$$

Two of these are state levels (λ, σ), one is the bounded Open-Gate source, and one is the constant rigidity of the old form. The decisive structural fact is that the index also depends on the *rate* of Sophia, hidden inside $\mu_1^0 = a_0 - c_0\lambda - c_1\dot{\lambda}^0$: through $\dot{\lambda}^0 = k_\sigma \sigma + I_{\mathrm{gate}} - \gamma\lambda$, the same σ and I_{gate} that raise the level also raise the rate.

Corollary 1.2 (Level and rate are the genuine drivers)**.** *Folding activation responds to the accumulated level λ* and *to the Sophia velocity $\dot{\lambda}$. The continuous/discrete distinction is not fundamental: what crosses the threshold is the combined level-plus-rate index A_{VFS}, however that increment is delivered.*

1.3 Level versus rate: rate-sensitivity of the threshold

If only the accumulated level mattered, the folding threshold would be the static value

$$\lambda_{\mathrm{fold}}^{(0)} = \frac{a_0}{c_0} \qquad (\dot{\lambda} = 0,\ \sigma = 0,\ I_{\mathrm{gate}} = 0). \tag{1.7}$$

Restoring the rate term shifts the effective threshold:

$$\boxed{\lambda_{\text{fold}}^{\text{eff}} = \frac{a_0 - c_1\dot{\lambda}}{c_0} \qquad \Longrightarrow \qquad \dot{\lambda} > 0 \;\Rightarrow\; \lambda_{\text{fold}}^{\text{eff}} < \lambda_{\text{fold}}^{(0)}.} \tag{1.8}$$

Proposition 1.2 (Rate can fold below the static level)**.** *There exist states with $\lambda < \lambda_{\text{fold}}^{(0)}$ and $A_{\text{VFS}} > 0$. Concretely, for $\sigma = I_{\text{gate}} = 0$ and growing Sophia $\dot{\lambda} > 0$, the old form is already unstable whenever*

$$c_0\lambda + c_1\dot{\lambda} > a_0, \tag{1.9}$$

which can hold at λ strictly below a_0/c_0 provided $\dot{\lambda} > (a_0 - c_0\lambda)/c_1 > 0$.

Proof. Immediate from $\mu_1^0 = a_0 - c_0\lambda - c_1\dot{\lambda} < 0$ with $\dot{\lambda}$ supplying the deficit $a_0 - c_0\lambda > 0$. □

This is the single most easily missed point: a surge of grace can fold a vessel that has not yet accumulated the static quota of Sophia. Velocity, not only stock, triggers the reconfiguration.

1.4 The two delivery channels

The same threshold is reached by two mechanisms, mirroring the parent corollary "grace is one but acts in two modes."

Modus I (continuous). Along a live arc, λ and σ evolve smoothly and A_{VFS} drifts. Folding triggers when the drift carries A_{VFS} through 0. This channel delivers *both* a rising level and a nonzero rate, so it is sensitive to Proposition 1.2.

Modus II (Resurrectio). The reset (1.2) injects a discontinuous increment of λ from the metabolised resistance, an impulsive analogue of $\dot{\lambda}$. It moves A_{VFS} along the very same axis, in one step.

The crucial point is that the reset does *not* reset q: the map (1.2) specifies only $\sigma, \lambda, \Omega_P, (V, F)$. Hence the shape amplitude is continuous across a reset, and only its target $q_* = \sqrt{Q_{\text{fold}}}$ jumps.

Corollary 1.3 (Resurrectio relocates the well; the shape relaxes)**.** *A Resurrectio event moves the shape potential $-\frac{1}{2}A_{\text{VFS}}q^2 + \frac{1}{4}B_{\text{VFS}}q^4$ instantaneously, after which the morphology relaxes by $\dot{q} = A_{\text{VFS}}^+ q - B_{\text{VFS}}^+ q^3$ toward the new minimum. Consequently:*

- *if a reset carries Q_{fold} from below 0 to above 0, the vessel dies unfolded and is reborn into a folding regime (*Resurrectio-triggered folding*);*
- *if a reset makes $Q_{\text{fold}} < 0$, an active fold is quenched, $q \to 0$.*

1.5 The Resurrectio jump in folding-space

We now compute the effect of one reset on the folding intensity. Write the metabolised resistance

$$m_R := (1 - q_R)\sigma^- \geq 0, \qquad \text{so} \qquad \sigma^+ = \sigma^- - m_R, \quad \lambda^+ = \lambda^- + m_R. \tag{1.10}$$

Remark (The metabolism conserves the balance sum). The (σ, λ) part of the reset satisfies $\sigma^+ + \lambda^+ = \sigma^- + \lambda^-$: it slides the state along the conservation line, converting resistance into Sophia without adding to the sum (the open balance constant jumps separately, through $\mathcal{G}_{\text{recepta}}$).

From (1.1),

$$\boxed{\Delta B_{\text{VFS}} = 2c_1\eta_{\text{fold}}\, m_R > 0, \qquad \Delta A_{\text{VFS}} = \Theta\, m_R + c_1 \Delta I_{\text{gate}}, \qquad \Theta := c_0 - c_1(\gamma + k_\sigma).} \tag{1.11}$$

The *folding-response coefficient* Θ decides whether converting one unit of resistance into Sophia raises ($\Theta > 0$) or lowers ($\Theta < 0$) the folding activation: gaining Sophia contributes $(c_0 - c_1\gamma)m_R$ (accumulation minus accelerated decay), while losing resistance removes $c_1 k_\sigma m_R$ of kathartic production. The gate correction $c_1\Delta I_{\text{gate}}$ has competing signs ($\sigma\downarrow$ opens the gate via $e^{-\phi\sigma}$; $\lambda\uparrow$ closes it via saturation) and is set aside in the structural results below.

Proposition 1.3 (Sign of the folding response and the critical level)**.** *Neglecting ΔI_{gate}, the jump in the folding intensity obeys*

$$\operatorname{sign} \Delta Q_{\text{fold}} = \operatorname{sign}\big(\Theta - 2c_1\eta_{\text{fold}}\, Q^-_{\text{fold}}\big), \tag{1.12}$$

so there is a universal critical level

$$\boxed{Q_{\text{crit}} = \frac{\Theta}{2c_1\eta_{\text{fold}}} = \frac{c_0 - c_1(\gamma + k_\sigma)}{2c_1\eta_{\text{fold}}}.} \tag{1.13}$$

Resurrectio raises folding when $Q^-_{\text{fold}} < Q_{\text{crit}}$ and lowers it when $Q^-_{\text{fold}} > Q_{\text{crit}}$. If $\Theta \le 0$ then $Q_{\text{crit}} \le 0$, so every reset strictly lowers Q_{fold} (the vessel unfolds at each death).

Proof. $\Delta Q_{\text{fold}} = \dfrac{B^-_{\text{VFS}}\Delta A_{\text{VFS}} - A^-_{\text{VFS}}\Delta B_{\text{VFS}}}{B^-_{\text{VFS}}(B^-_{\text{VFS}} + \Delta B_{\text{VFS}})}$. The denominator is positive; the numerator is $B^-_{\text{VFS}}\Theta m_R - A^-_{\text{VFS}} 2c_1\eta_{\text{fold}} m_R = B^-_{\text{VFS}} m_R(\Theta - 2c_1\eta_{\text{fold}} Q^-_{\text{fold}})$ using $A^-_{\text{VFS}} = Q^-_{\text{fold}} B^-_{\text{VFS}}$. Hence the stated sign and fixed level. □

Proposition 1.4 (Each reset is an exact contraction toward Q_{crit})**.** *Neglecting ΔI_{gate}, the jump map on the folding intensity is affine-over-affine,*

$$Q^+_{\text{fold}} = \frac{Q^-_{\text{fold}} B^-_{\text{VFS}} + \Theta m_R}{B^-_{\text{VFS}} + 2c_1\eta_{\text{fold}} m_R}, \tag{1.14}$$

and satisfies the exact contraction identity

$$\boxed{Q^+_{\text{fold}} - Q_{\text{crit}} = \kappa_{\text{fold}}\,\big(Q^-_{\text{fold}} - Q_{\text{crit}}\big), \qquad \kappa_{\text{fold}} = \frac{B^-_{\text{VFS}}}{B^-_{\text{VFS}} + 2c_1\eta_{\text{fold}} m_R} \in (0,1).} \tag{1.15}$$

The fixed point Q_{crit} is independent of m_R, of B^-_{VFS}, and of the depth of death.

Proof. Direct substitution of $Q_{\text{crit}} = \Theta/(2c_1\eta_{\text{fold}})$:

$$Q^+_{\text{fold}} - Q_{\text{crit}} = \frac{2c_1\eta_{\text{fold}}(Q^-_{\text{fold}} B^-_{\text{VFS}} + \Theta m_R) - \Theta(B^-_{\text{VFS}} + 2c_1\eta_{\text{fold}} m_R)}{2c_1\eta_{\text{fold}}(B^-_{\text{VFS}} + 2c_1\eta_{\text{fold}} m_R)} = \frac{B^-_{\text{VFS}}(2c_1\eta_{\text{fold}} Q^-_{\text{fold}} - \Theta)}{2c_1\eta_{\text{fold}}(B^-_{\text{VFS}} + 2c_1\eta_{\text{fold}} m_R)},$$

which equals $\kappa_{\text{fold}}(Q^-_{\text{fold}} - Q_{\text{crit}})$. □

Proposition 1.5 (Conditional convergence over a reset sequence). *For an admissible sequence of resets with metabolised increments $m_R^{(j)}$ and pre-jump denominators $B_{\rm VFS}^{(j)}$,*

$$Q_{\rm fold}^{(N)} - Q_{\rm crit} = \Big(\prod_{j=1}^{N} \kappa_{\rm fold}^{(j)}\Big)\big(Q_{\rm fold}^{(0)} - Q_{\rm crit}\big), \qquad Q_{\rm fold} \to Q_{\rm crit} \iff \sum_j m_R^{(j)} = \infty. \quad (1.16)$$

Proof. Telescoping Proposition 1.4. Since $1-\kappa_{\rm fold}^{(j)} = 2c_1\eta_{\rm fold}m_R^{(j)}/(B_{\rm VFS}^{(j)}+2c_1\eta_{\rm fold}m_R^{(j)})$, the product $\prod \kappa_{\rm fold}^{(j)} \to 0$ iff $\sum(1-\kappa_{\rm fold}^{(j)}) = \infty$, which (with $B_{\rm VFS}^{(j)}$ bounded below) is equivalent to $\sum m_R^{(j)} = \infty$. □

If the resistance budget is exhausted ($\sum m_R < \infty$, e.g. σ depleted with no replenishment), repeated resurrections converge to a point strictly short of $Q_{\rm crit}$. Reaching the universal level requires sustained replenishment, $\sum m_R = \infty$ — the folding analogue of the expansion-budget condition $\mathcal{G}_{\rm recepta}(\infty) = +\infty$ in the cosmological layer.

Remark (Gate correction). Restoring $\Delta I_{\rm gate}$ adds $c_1\Delta I_{\rm gate}$ to the numerator of $\Delta A_{\rm VFS}$, shifting the effective fixed point to $Q'_{\rm crit} = Q_{\rm crit} + \Delta I_{\rm gate}/(2\eta_{\rm fold}m_R)$; its sign is state-dependent. The existence of a contracting fixed point is unchanged.

Corollary 1.4 (Chart-regularity of repeated resurrection). *If $0 < Q_{\rm crit} < q_{\rm max}^2$, the contraction of Proposition 1.4 keeps the folded amplitude inside the regular one-mode chart under any admissible reset sequence. This is a folding companion to the grace-window condition* $\max\{R_c, R_{\rm min}^{\rm ceil}\} \le \mathcal{G}^-_{\rm recepta} \le R_{\rm max}$*: resurrection is admissible* and *chart-regular when both hold.*

1.6 Curvature-pocket promotion by Resurrectio

The folded scalar curvature is $R_{\rm folded} = -2/\Omega_P^2 + \rho_1 Q_{\rm fold}$, with positive-pocket threshold $q_{\rm curv}^2 = 2/(\rho_1\Omega_P^2)$. Across a reset both terms move:

$$\boxed{\Delta R_{\rm folded} = \underbrace{2\left[\frac{1}{(\Omega_P^-)^2} - \frac{1}{(\Omega_P^+)^2}\right]}_{>0:\text{ background release}} + \rho_1\,\Delta Q_{\rm fold}.} \quad (1.17)$$

Simultaneously the pocket threshold falls, because Ω_P grows:

$$q_{\rm curv}^{2,+} = \frac{2}{\rho_1(\Omega_P^+)^2} < \frac{2}{\rho_1(\Omega_P^-)^2} = q_{\rm curv}^{2,-}. \quad (1.18)$$

Proposition 1.6 (Resurrectio is pocket-promoting in the sub-critical regime). *If $\rho_1 > 0$, $\Theta > 0$, and $Q_{\rm fold}^- < Q_{\rm crit}$, then a single reset both raises the folded amplitude ($\Delta Q_{\rm fold} > 0$) and lowers the pocket threshold ($q_{\rm curv}^{2,+} < q_{\rm curv}^{2,-}$). Both effects push $R_{\rm folded}$ upward, so a death-and-resurrection event can convert a smooth hyperbolic patch into a positive-curvature pocket.*

Proof. $\Delta Q_{\rm fold} > 0$ by Proposition 1.3 under $Q_{\rm fold}^- < Q_{\rm crit}$, $\Theta > 0$; the threshold drop follows from $\Omega_P^+ > \Omega_P^-$. Both terms of $\Delta R_{\rm folded}$ are then positive, and the pocket criterion $Q_{\rm fold} > q_{\rm curv}^2$ is approached from both sides. □

Corollary 1.5 (Resurrection concentrates form)**.** *The continuous mode (Epektasis) relaxes curvature, $\dot{R}_\Sigma \geq 0$ with $R \to 0^-$; the discrete mode can locally* concentrate *it. Resurrectio does not only enlarge the Brim — in the sub-critical folding regime it gives the renewed vessel a more sharply formed morphology.*

1.7 The energy reading corrected

It is tempting to read folding as the surface being forced into a "less efficient" form. Within the formalism this is inverted: the folded form carries *lower* morphological tension. With $\mathcal{T}_{\text{fold}}(q) = -\frac{1}{2}A_{\text{VFS}}q^2 + \frac{1}{4}B_{\text{VFS}}q^4$,

$$\mathcal{T}_{\text{fold}}(0) = 0, \qquad \mathcal{T}_{\text{fold}}(q_*) = -\frac{A_{\text{VFS}}^2}{4B_{\text{VFS}}} < 0 \qquad (A_{\text{VFS}} > 0), \tag{1.19}$$

so the resolved tension is $\Delta\mathcal{T}_{\text{fold}} = A_{\text{VFS}}^2/(4B_{\text{VFS}}) > 0$ and the folded branch is energetically preferred. The reconfiguration is toward a better-adapted state, not a worse one.

The correct sense in which the old form is "inefficient" is as a *carrier* of Sophia past the threshold: it cannot absorb the Sophianic pressure, so tension builds until the symmetric branch destabilises. The folded form is the efficient carrier — it embodies the excess through the feedback $-\eta_{\text{fold}}q^2\lambda$, raising the effective damping

$$\gamma_{\text{eff}} = \gamma + \eta_{\text{fold}}Q_{\text{fold}} > \gamma, \qquad \lambda^*_{\text{fold}} = \frac{J_{\text{in}}}{\gamma + \eta_{\text{fold}}q^2} < \frac{J_{\text{in}}}{\gamma} = \lambda^*_{\text{old}}, \tag{1.20}$$

and thereby lowering the level of unembodied Sophia.

The surface is not forced into a worse form. The old form stops being an adequate vessel for the attained level/rate of Sophia, loses stability, and the morphology relaxes into a lower-tension folded form that carries the Sophia better.

Compact summary

$$\boxed{\text{single trigger: } A_{\text{VFS}} = 0 \ (\mu_1^0 = 0) \text{ — supercritical pitchfork, } q = 0 \to q_\pm.}$$

$$\boxed{\begin{gathered} A_{\text{VFS}} = (c_0 - c_1\gamma)\lambda + c_1 k_\sigma \sigma + c_1 I_{\text{gate}} - a_0, \\ \text{rate enters via } \mu_1^0 = a_0 - c_0\lambda - c_1\dot{\lambda}^0. \end{gathered}}$$

$$\boxed{\begin{gathered} \text{level } \lambda \text{ and rate } \dot{\lambda} \text{ are the drivers;} \\ \text{Modus I (continuous) and Modus II (Resurrectio) are channels.} \end{gathered}}$$

$$\boxed{\Theta = c_0 - c_1(\gamma + k_\sigma), \qquad Q_{\text{crit}} = \frac{\Theta}{2c_1\eta_{\text{fold}}}, \qquad Q^+_{\text{fold}} - Q_{\text{crit}} = \kappa_{\text{fold}}(Q^-_{\text{fold}} - Q_{\text{crit}}),\ \kappa_{\text{fold}} \in (0,1).}$$

$$\boxed{Q_{\text{fold}} \to Q_{\text{crit}} \iff \sum_j m_R^{(j)} = \infty; \qquad \text{chart-regular if } 0 < Q_{\text{crit}} < q^2_{\text{max}}.}$$

$$\boxed{\begin{array}{c}\text{sub-critical reset is pocket-promoting;}\\ \text{folded form has lower tension } \mathcal{T}_{\mathrm{fold}}(q_*) = -A_{\mathrm{VFS}}^2/4B_{\mathrm{VFS}} < 0.\end{array}}$$

V.F.S. v2.0 (Open-Gate) · Folding-Trigger Layer. Derived from the shape potential and Sophia-production law of `vfs_geometry.tex`, the Brim/curvature data of the geometric and cosmological layers, and the Resurrectio reset of `vfs_opengate_lyapunov.tex`. Propositions are closed-form; corollaries are interpretive. Folding parameters are not fixed by the source files; numerical levels are illustrative.

Chapter 2

Folding–Lyapunov Addendum

The Sophianic Folding extension of the Open-Gate Lyapunov file establishes the continuous-arc compatibility $\dot{\mathcal{L}}_{\text{ext}} \le C_{\text{ext}} - C_1\mathcal{L}_{\text{ext}}$ under an adiabatic hypothesis, but leaves three items asserted rather than derived: the positive invariance of the regular folding chart, the behaviour of the folding functional across the Resurrectio reset, and the choice between two distinct activation-index conventions. This addendum closes them. It first records a correction — the file's folding functional $\mathcal{V}_{\text{fold}} = \frac{B_{\text{fold}}}{4}(q_{\text{fold}}^2 - Q_{\text{fold}}^*)^2$ is unbounded *on the epektasis domain, because $B_{\text{fold}} = b + 2c_1\eta_{\text{fold}}\lambda \to \infty$; the B_{fold}-normalised functional $\widetilde{\mathcal{V}}_{\text{fold}} = \mathcal{V}_{\text{fold}}/B_{\text{fold}}$ is bounded and equally descending. It then proves (i) Nagumo invariance of $\mathcal{D}_A^{\text{fold}} = \mathcal{D}_A \times [0, q_{\max}]$ under the standing folding wall $Q_{\text{fold}}^* \le q_{\max}^2$, (ii) a folding reset bound: since $\mathfrak{R}$ does not act on q_{fold}, the chart is preserved across deaths and $\widetilde{\mathcal{L}}_{\text{ext}}(x^+)$ stays bounded, whence the hybrid trajectory is forward complete for $\widetilde{\mathcal{L}}_{\text{ext}}$; and (iii) a reconciliation declaring the core-consistent Sophia rate canonical, with the geometry-layer parameters as an effective surrogate. The invariance and reset results are convention-independent; only the closed-form critical level Q_{crit} is convention-specific.*

2.1 Imported objects and the three open items

From the folding extension of the Lyapunov file we import the shape amplitude q_{fold}, its intensity $Q_{\text{fold}} = q_{\text{fold}}^2$, the activation index A_{fold}, the quartic coefficient $B_{\text{fold}} = b + 2c_1\eta_{\text{fold}}\lambda > 0$, the one-mode gradient flow

$$\dot{q_{\text{fold}}} = A_{\text{fold}}q_{\text{fold}} - B_{\text{fold}}q_{\text{fold}}^3 = -B_{\text{fold}}q_{\text{fold}}\big(q_{\text{fold}}^2 - Q_{\text{fold}}^*\big), \qquad Q_{\text{fold}}^* := \frac{A_{\text{fold}}}{B_{\text{fold}}}, \tag{2.1}$$

and the base certificate $\dot{\mathcal{L}_{\text{VFS}}} \le C - C_1\mathcal{L}_{\text{VFS}}$ on the positively-invariant active domain $\mathcal{D}_A$, with the non-Zeno dwell-time bound $\Delta t_j \ge \Delta t_* > 0$ and the reset map

$$\mathfrak{R}: \quad \sigma^+ = q_R\sigma^-, \quad \lambda^+ = \lambda^- + (1-q_R)\sigma^-, \quad (V,F) \text{ continuous}, \quad \Omega_P^+ = \Omega_P^- + \kappa_R\mathcal{G}_{\text{recepta}}^-, \tag{2.2}$$

with $0 \le q_R < 1$, $\kappa_R > 0$, $u^+ = u^-$. The file proves $x^- \in \mathcal{D}_A \Rightarrow x^+ \in \mathcal{D}_A$ in the core coordinates and $\mathcal{L}_{\text{VFS}}(x^+) \le L_R := \sup_{\mathcal{D}_A}\mathcal{L}_{\text{VFS}}$.

The three items left open are: (i) the regular folding domain $0 \le Q_{\text{fold}} \le q_{\max}^2$ is *assumed*, not shown invariant; (ii) the reset bound and forward completeness are stated for $\mathcal{L}_{\text{VFS}}$, while the behaviour of the folding functional across $\mathfrak{R}$ is not controlled; (iii) the activation index is written with the core Sophia rate here but with a different, minimal rate in the geometric and cosmological layers.

2.2 Correction: a bounded folding functional

The file shifts the folding potential to a non-negative functional

$$\mathcal{V}_{\text{fold}} = \frac{B_{\text{fold}}}{4}\Big(q_{\text{fold}}^2 - Q_{\text{fold}}^*\Big)^2 \ge 0 \qquad (A_{\text{fold}} > 0), \tag{2.3}$$

and claims it is bounded above by a constant M_{fold} on the regular domain. This fails on the Open-Gate domain: epektasis drives $\Omega_P \to \infty$ and λ is not bounded on $\mathcal{D}_A$ ($\lambda \le q^*\Omega_P$), so $B_{\text{fold}} = b + 2c_1\eta_{\text{fold}}\lambda \to \infty$ and $\mathcal{V}_{\text{fold}}$ is unbounded even at fixed $|q_{\text{fold}}^2 - Q_{\text{fold}}^*|$. The reset bound $\mathcal{L}_{\text{ext}}(x^+) \le L_R + w_{\text{fold}}M_{\text{fold}}$ therefore does not close as written.

The remedy is to use the B_{fold}-normalised functional

$$\boxed{\widetilde{\mathcal{V}}_{\text{fold}} := \frac{\mathcal{V}_{\text{fold}}}{B_{\text{fold}}} = \frac{1}{4}\Big(q_{\text{fold}}^2 - Q_{\text{fold}}^*\Big)^2 \ge 0, \qquad Q_{\text{fold}}^* = \frac{A_{\text{fold}}}{B_{\text{fold}}} \text{ (signed).}} \tag{2.4}$$

This requires no sign split: it is non-negative for either sign of A_{fold}, and it descends along (2.1).

Lemma 2.1 (Descent of the bounded functional)**.** *Along* (2.1) *with frozen* Q_{fold}^**,*

$$\boxed{\dot{\widetilde{\mathcal{V}}}_{\text{fold}} = -B_{\text{fold}}\, q_{\text{fold}}^2\big(q_{\text{fold}}^2 - Q_{\text{fold}}^*\big)^2 \le 0.} \tag{2.5}$$

For varying parameters, $\dot{\widetilde{\mathcal{V}}}_{\text{fold}} = -B_{\text{fold}}q_{\text{fold}}^2(q_{\text{fold}}^2 - Q_{\text{fold}}^*)^2 - \frac{1}{2}(q_{\text{fold}}^2 - Q_{\text{fold}}^*)\dot{Q}_{\text{fold}}^*$*, and the remainder is bounded on any set with* $q_{\text{fold}} \in [0, q_{\max}]$*,* $|Q_{\text{fold}}^*|$ *and* $|\dot{Q}_{\text{fold}}^*|$ *bounded.*

Proof. Using $A_{\text{fold}} = B_{\text{fold}}Q_{\text{fold}}^*$ in (2.1), $\dot{q}_{\text{fold}} = -B_{\text{fold}}q_{\text{fold}}(q_{\text{fold}}^2 - Q_{\text{fold}}^*)$. With frozen Q_{fold}^*, $\dot{\widetilde{\mathcal{V}}}_{\text{fold}} = \frac{1}{2}(q_{\text{fold}}^2 - Q_{\text{fold}}^*)(2q_{\text{fold}}\dot{q}_{\text{fold}}) = (q_{\text{fold}}^2 - Q_{\text{fold}}^*)q_{\text{fold}}\dot{q}_{\text{fold}} = -B_{\text{fold}}q_{\text{fold}}^2(q_{\text{fold}}^2 - Q_{\text{fold}}^*)^2 \le 0$. For moving Q_{fold}^* the extra term is $-\frac{1}{2}(q_{\text{fold}}^2 - Q_{\text{fold}}^*)\dot{Q}_{\text{fold}}^*$, with $|\frac{1}{2}(q_{\text{fold}}^2 - Q_{\text{fold}}^*)\dot{Q}_{\text{fold}}^*| \le \frac{1}{2}(q_{\max}^2 + |Q_{\text{fold}}^*|)\,|\dot{Q}_{\text{fold}}^*|$. ☐

$\widetilde{\mathcal{V}}_{\text{fold}} = \mathcal{V}_{\text{fold}}/B_{\text{fold}}$ *is the same Lyapunov object re-weighted by the positive function* B_{fold}*; re-weighting preserves the descent property, and the normalisation removes the only source of unboundedness. From here the extended certificate is* $\widetilde{\mathcal{L}}_{\text{ext}} := \mathcal{L}_{\text{VFS}} + w_{\text{fold}}\widetilde{\mathcal{V}}_{\text{fold}}$*,* $w_{\text{fold}} > 0$*.*

2.3 The standing folding wall and its finiteness

Standing Hypothesis 2.1 (Folding wall)**.** The chart half-width $q_{\max}$ satisfies

$$\boxed{q_{\max}^2 \;\ge\; Q^{*,\sup} := \sup_{\mathcal{D}_A} Q_{\text{fold}}^*.} \tag{2.6}$$

Lemma 2.2 (The wall is attainable)**.** *In the geometric (minimal-rate) convention* $A_{\text{fold}} = (c_0 - c_1\gamma)\lambda + c_1k_\sigma\sigma + c_1I_{\text{gate}} - a_0$*,* $B_{\text{fold}} = b + 2c_1\eta_{\text{fold}}\lambda$*, the supremum is finite and explicit:*

$$Q^{*,\sup} = \max\left\{\frac{c_1(k_\sigma C_\sigma + \zeta_0) - a_0}{b},\ \frac{c_0 - c_1\gamma}{2c_1\eta_{\text{fold}}}\right\}. \tag{2.7}$$

Proof. For fixed $(\sigma, I_{\text{gate}})$, $Q_{\text{fold}}^*(\lambda)$ is a ratio of two affine functions of $\lambda \ge 0$, hence monotone, with horizontal asymptote $\lim_{\lambda\to\infty} Q_{\text{fold}}^* = (c_0 - c_1\gamma)/(2c_1\eta_{\text{fold}})$ (independent of σ, I_{gate}). Its supremum over $\lambda \ge 0$ is attained at an endpoint: at $\lambda = 0$, $Q_{\text{fold}}^* = (c_1k_\sigma\sigma + c_1I_{\text{gate}} - a_0)/b$, maximised over the box $\sigma \le C_\sigma$, $I_{\text{gate}} \le \zeta_0$ at the stated value; or at $\lambda \to \infty$, the asymptote. Both are finite, so $Q^{*,\sup} < \infty$. ☐

If $Q^{*,\sup} \le 0$ *the old form is everywhere stable and any* $q_{\max} > 0$ *satisfies the wall. The non-trivial case is* $Q^{*,\sup} > 0$*, where Hypothesis 2.1 fixes a minimal admissible chart. The canonical (core-rate) convention saturates likewise:* $Q_{\text{fold}}^* \to c_0/(2c_1\eta_{\text{fold}})$ *under cleansing* $\sigma \to 0$*, since* $(\delta u - \gamma)\tanh(\kappa\sigma) \to 0$ *even as* $u \to \infty$ *(§6).*

2.4 Folding-chart invariance (Nagumo)

Definition 2.1 (Extended active domain)**.** $\mathcal{D}_A^{\text{fold}} := \mathcal{D}_A \times [0, q_{\max}]$, adjoining the folding coordinate q_{fold} to the core active domain.

Lemma 2.3 (Folding-face sub-tangentiality)**.** *Under Hypothesis 2.1, the flow* (2.1) *is sub-tangential on the folding faces of* $\mathcal{D}_A^{\text{fold}}$*:*

$$\dot{q}_{\text{fold}}\big|_{q_{\text{fold}}=0} = 0 \quad (\textit{tangent}), \qquad \dot{q}_{\text{fold}}\big|_{q_{\text{fold}}=q_{\max}} = -B_{\text{fold}}\, q_{\max}\big(q_{\max}^2 - Q_{\text{fold}}^*\big) \le 0 \quad (\textit{inward}). \tag{2.8}$$

Consequently $[0, q_{\max}]$ *is forward invariant for* q_{fold}*, and* $\mathcal{D}_A^{\text{fold}}$ *is positively invariant under the continuous flow.*

Proof. At $q_{\text{fold}} = 0$, (2.1) gives $\dot{q}_{\text{fold}} = 0$, so $\{q_{\text{fold}} \geq 0\}$ is preserved. At $q_{\text{fold}} = q_{\max}$, $\dot{q}_{\text{fold}} = -B_{\text{fold}} q_{\max}(q_{\max}^2 - Q_{\text{fold}}^*)$ with $B_{\text{fold}} > 0$, $q_{\max} > 0$, and $q_{\max}^2 - Q_{\text{fold}}^* \geq 0$ by Hypothesis 2.1; hence $\dot{q}_{\text{fold}} \leq 0$. All core faces are sub-tangential by the base Nagumo lemma of the Lyapunov file, and the two folding faces are sub-tangential here; therefore $\mathcal{D}_A^{\text{fold}}$ is positively invariant. □

2.5 The folding reset bound and hybrid completeness

Lemma 2.4 (Folding reset bound)**.** *The reset* (2.2) *does not act on* q_{fold}*; hence* $q_{\text{fold}}^+ = q_{\text{fold}}^-$*. Under Hypotheses of the base reset lemma together with Hypothesis 2.1:*

(1) $x^- \in \mathcal{D}_A^{\text{fold}} \;\Rightarrow\; x^+ \in \mathcal{D}_A^{\text{fold}}$*, because* $x^+ \in \mathcal{D}_A$ *(base reset lemma) and* $q_{\text{fold}}^+ = q_{\text{fold}}^- \in [0, q_{\max}]$*;*

(2) $\widetilde{\mathcal{V}}_{\text{fold}}$ *is bounded on* $\mathcal{D}_A^{\text{fold}}$*,*

$$\widetilde{\mathcal{V}}_{\text{fold}} \leq \widetilde{M}_{\text{fold}} := \tfrac{1}{4} \max\bigl(q_{\max}^2 - Q_{\min}^*,\ |Q_{\max}^*|\bigr)^2 < \infty, \tag{2.9}$$

so the extended certificate satisfies

$$\boxed{\widetilde{\mathcal{L}}_{\text{ext}}(x^+) \leq \widetilde{L}_R := L_R + w_{\text{fold}} \widetilde{M}_{\text{fold}} < \infty.} \tag{2.10}$$

Proof. (1) The map (2.2) specifies only $\sigma, \lambda, (V, F), \Omega_P$; q_{fold} is inert, so $q_{\text{fold}}^+ = q_{\text{fold}}^-$, which lies in $[0, q_{\max}]$ if q_{fold}^- did. With $x^+ \in \mathcal{D}_A$ from the base lemma, $x^+ \in \mathcal{D}_A^{\text{fold}}$. (2) On $\mathcal{D}_A^{\text{fold}}$, $q_{\text{fold}}^2 \in [0, q_{\max}^2]$ and $Q_{\text{fold}}^* \in [Q_{\min}^*, Q_{\max}^*]$ bounded (Lemma 2.2), so $\widetilde{\mathcal{V}}_{\text{fold}} = \frac{1}{4}(q_{\text{fold}}^2 - Q_{\text{fold}}^*)^2 \leq \widetilde{M}_{\text{fold}} < \infty$. Adding $w_{\text{fold}} \widetilde{\mathcal{V}}_{\text{fold}}$ to $\mathcal{L}_{\text{VFS}}(x^+) \leq L_R$ gives the bound. □

Proposition 2.1 (Hybrid forward completeness for $\widetilde{\mathcal{L}}_{\text{ext}}$)**.** *Assume the regular folding conditions: Hypothesis 2.1,* $B_{\text{fold}} \geq B_{\min} > 0$*, and* $|Q_{\text{fold}}^*|$ *bounded on* $\mathcal{D}_A^{\text{fold}}$ *(adiabatic regularity). Then there is a finite* $\widetilde{C}_{\text{ext}}$ *with*

$$\dot{\widetilde{\mathcal{L}}}_{\text{ext}} \leq \widetilde{C}_{\text{ext}} - C_1 \widetilde{\mathcal{L}}_{\text{ext}} \qquad \textit{on continuous arcs}, \tag{2.11}$$

and the same contraction rate C_1 *as the base certificate. Combined with the reset bound (Lemma 2.4) and the unchanged non-Zeno dwell time, the hybrid trajectory satisfies*

$$\boxed{\widetilde{\mathcal{L}}_{\text{ext}}(t) \leq \widetilde{L}_* := \max\Bigl\{\widetilde{L}_R,\ \widetilde{C}_{\text{ext}}/C_1\Bigr\}} \tag{2.12}$$

across any infinite admissible reset sequence; resets do not accumulate, and the trajectory is forward complete in the folding-extended state.

Proof. By Lemma 2.1 and the base estimate, $\dot{\widetilde{\mathcal{L}}}_{\text{ext}} = \dot{\mathcal{L}}_{\text{VFS}} + w_{\text{fold}} \dot{\widetilde{\mathcal{V}}}_{\text{fold}} \leq (C - C_1 \mathcal{L}_{\text{VFS}}) + w_{\text{fold}}\bigl(-B_{\text{fold}} q_{\text{fold}}^2 (q_{\text{fold}}^2 - Q_{\text{fold}}^*)^2 + R\bigr)$ with $|R| \leq R_{\text{fold}} < \infty$ the bounded adiabatic remainder (Lemma 2.1). Dropping the non-positive descent term and writing $\mathcal{L}_{\text{VFS}} = \widetilde{\mathcal{L}}_{\text{ext}} - w_{\text{fold}} \widetilde{\mathcal{V}}_{\text{fold}} \geq \widetilde{\mathcal{L}}_{\text{ext}} - w_{\text{fold}} \widetilde{M}_{\text{fold}}$,

$$\dot{\widetilde{\mathcal{L}}}_{\text{ext}} \leq C - C_1 \widetilde{\mathcal{L}}_{\text{ext}} + C_1 w_{\text{fold}} \widetilde{M}_{\text{fold}} + w_{\text{fold}} R_{\text{fold}} =: \widetilde{C}_{\text{ext}} - C_1 \widetilde{\mathcal{L}}_{\text{ext}}, \tag{2.13}$$

with $\widetilde{C}_{\text{ext}} < \infty$ now genuinely finite (it uses $\widetilde{M}_{\text{fold}}$, not the unbounded M_{fold}). On each arc $\widetilde{\mathcal{L}}_{\text{ext}}(t) \leq \max\{\widetilde{\mathcal{L}}_{\text{ext}}(t_j^+), \widetilde{C}_{\text{ext}}/C_1\}$; resets give $\widetilde{\mathcal{L}}_{\text{ext}}(x^+) \leq \widetilde{L}_R$ (Lemma 2.4); the dwell time is unchanged, since q_{fold} adds no reset-triggering boundary (its chart face is invariant-inward by Lemma 2.3, not a death face) and resets are exogenous on the core coordinates. Hence $\widetilde{\mathcal{L}}_{\text{ext}} \leq \widetilde{L}_*$ for all t and the hybrid trajectory is forward complete. □

The folding velocity $\dot{q}_{\text{fold}}$ *may grow with* λ *off the slow manifold* $q_{\text{fold}} = \sqrt{Q_{\text{fold}}^*}$*, but this is a strictly inward restoring rate; it neither triggers nor delays resets, which are exogenous and defined on the core coordinates. The non-Zeno bound is therefore inherited from the base field bound* $\bar{M}_*$*, unchanged.*

2.6 Reconciliation of the two activation-index conventions

Two definitions of the activation index appear across the manuscript:

$$\text{Canonical (core rate):}\quad A_{\text{fold}} = c_0\lambda + c_1\dot\lambda^{(0)} - a_0, \quad \dot\lambda^{(0)} = (\delta u - \gamma)\tanh(\kappa\sigma) + I_{\text{gate}}; \tag{2.14}$$

$$\text{Geometric (minimal rate):}\quad A_{\text{fold}} = (c_0 - c_1\gamma)\lambda + c_1 k_\sigma\sigma + c_1 I_{\text{gate}} - a_0, \tag{2.15}$$

the latter from the surrogate production law $\dot\lambda = k_\sigma\sigma + I_{\text{gate}} - \gamma\lambda - \eta_{\text{fold}} q_{\text{fold}}^2\lambda$. Since λ already has a defined production law in the core, we take the **canonical** form as primary and read the geometric parameters as an effective linear surrogate near an operating point $(\bar u, \bar\sigma)$:

$$\boxed{c_1\big(\delta\bar u - \gamma\big)\tanh(\kappa\sigma) \;\longleftrightarrow\; c_1 k_\sigma\sigma - c_1\gamma\lambda, \quad \text{i.e. } k_\sigma \approx \kappa(\delta\bar u - \gamma),\;\; \gamma \text{ absorbing the linear } \lambda\text{-decay.}} \tag{2.16}$$

This surrogate is structurally imperfect (a σ-term plus a λ-decay standing in for a single $u\tanh\sigma$ term), so the two conventions are genuinely distinct models, not exact reparametrisations.

Proposition 2.2 (What is and is not convention-dependent)**.** *Lemmas 2.1, 2.3, 2.4 and Proposition 2.1 use only: the gradient form* (2.1)*, the wall $Q^*_{\text{fold}} \le q^2_{\max}$, the inertness of q_{fold} under $\mathfrak{R}$, and the boundedness of $Q^*_{\text{fold}}, \dot Q^*_{\text{fold}}$. None of these uses the specific form of A_{fold}. Hence the invariance, reset bound, and hybrid completeness are* convention-independent. *Only the closed-form critical level and contraction of the reset map are convention-specific.*

Corollary 2.1 (Critical level by convention)**.** *Write the metabolised resistance $m_R = (1 - q_R)\sigma^-$, so $\sigma^+ = \sigma^- - m_R$, $\lambda^+ = \lambda^- + m_R$. In the* geometric *convention the reset acts on the target as an exact contraction toward a constant critical level,*

$$\Theta = c_0 - c_1(\gamma + k_\sigma), \qquad Q_{\text{crit}} = \frac{\Theta}{2c_1\eta_{\text{fold}}}, \qquad {Q^*_{\text{fold}}}^+ - Q_{\text{crit}} = \frac{B^-_{\text{fold}}}{B^-_{\text{fold}} + 2c_1\eta_{\text{fold}} m_R}\big({Q^*_{\text{fold}}}^- - Q_{\text{crit}}\big), \tag{2.17}$$

recovering the result of the folding-trigger note. In the canonical *convention the metabolism changes A_{fold} by $c_0 m_R + c_1(\delta u - \gamma)\big[\tanh(\kappa\sigma^+) - \tanh(\kappa\sigma^-)\big]$, which is not affine in m_R; the same qualitative contraction holds, but toward a state-dependent effective level $Q_{\text{crit}}(t)$ rather than a constant.*

Remark (Chart-regularity of repeated resurrection). Combining Corollary 2.1 with Hypothesis 2.1: in the geometric convention, if $0 < Q_{\text{crit}} < q^2_{\max}$ and $Q^{*,\sup} \le q^2_{\max}$, then the reset contraction keeps the target inside the chart at every death, so the regular folding domain is preserved under any admissible reset sequence. This is the folding companion to the grace-window admissibility $\max\{R_c, R^{\text{ceil}}_{\min}\} \le \mathcal{G}^-_{\text{recepta}} \le R_{\max}$: resurrection is admissible *and* chart-regular when both windows hold.

2.7 Compact summary

$$\boxed{\widetilde{\mathcal{V}}_{\text{fold}} = \tfrac14 (q^2_{\text{fold}} - Q^*_{\text{fold}})^2 = \mathcal{V}_{\text{fold}}/B_{\text{fold}} \text{ bounded on epektasis;} \qquad \dot{\widetilde{\mathcal{V}}}_{\text{fold}} = -B_{\text{fold}} q^2_{\text{fold}}(q^2_{\text{fold}} - Q^*_{\text{fold}})^2 \le 0.}$$

$$\boxed{\text{wall } q^2_{\max} \ge Q^{*,\sup} < \infty \;\Rightarrow\; \mathcal{D}^{\text{fold}}_A = \mathcal{D}_A \times [0, q_{\max}] \text{ invariant (Nagumo).}}$$

$$\boxed{\mathfrak{R} \text{ inert on } q_{\text{fold}} \;\Rightarrow\; x^+ \in \mathcal{D}^{\text{fold}}_A, \quad \widetilde{\mathcal{L}}_{\text{ext}}(x^+) \le \widetilde{L}_R < \infty.}$$

$$\boxed{\dot{\widetilde{\mathcal{L}}}_{\text{ext}} \le \widetilde{C}_{\text{ext}} - C_1\widetilde{\mathcal{L}}_{\text{ext}}, \quad \widetilde{\mathcal{L}}_{\text{ext}}(t) \le \widetilde{L}_*; \quad \text{forward complete, non-Zeno inherited.}}$$

$$\boxed{\text{invariance + reset bound are convention-independent; } Q_{\text{crit}} \text{ is convention-specific.}}$$

V.F.S. v2.0 (Open-Gate) · Folding–Lyapunov Addendum. Closes the invariance, reset-bound, and convention items left open in the Sophianic Folding extension of `vfs_opengate_lyapunov.tex`*, and corrects the folding functional to a bounded B_{fold}-normalised form. Results are domain-conditional on the positively-invariant $\mathcal{D}^{\text{fold}}_A$; the conclusion is set stability with partial convergence, now including morphology, not global or automatic stability.*

Part II

Structure Formation on the Vessel

Chapter 3

Structure Formation on the Vessel

Scope and epistemic status

Every layer so far has been spatially *homogeneous*: the Vessel is a $2{+}1$ open FLRW background $a(t) = \Omega_P(t)$ with a time-only Sophia field $\lambda(t)$. This note opens the inhomogeneous sector — how local concentrations of Sophia and form grow or decay on the expanding hyperbolic surface. As in the parent files, *Propositions* are closed-form results and *Corollaries* are interpretive. The analysis is at the linear, test-field level on a fixed background; metric back-reaction is a named next layer. Two transport constants (D for Sophia, D_q for form) are introduced; they are *new* modelling inputs, absent from the base dynamical system, and are flagged as such.

The background is

$$ds^2 = -dt^2 + \Omega_P(t)^2\, h_{ij}\, dx^i dx^j, \qquad K_h = -1, \qquad H = \frac{\dot{\Omega}_P}{\Omega_P} = \frac{\alpha\lambda}{\Omega_P}, \qquad \dot{\Omega}_P = \alpha\lambda, \tag{3.1}$$

with h the metric of the hyperbolic spatial surface $\mathbb{H}^2$.

3.1 The relaxational thesis: which field can carry structure

In standard cosmology, density structure grows by gravitational instability because the matter field carries inertia: a second-order-in-time wave equation with a sound speed permits oscillation and Jeans-unstable growth. The Sophia field of V.F.S. is different in kind. Its production law is *first order in time*,

$$\dot{\lambda} = (\delta u - \gamma)\tanh(\kappa\sigma) + I_{\text{gate}} - \eta_{\text{fold}} q_{\text{fold}}^2 \lambda, \tag{3.2}$$

a relaxation (reaction) law with no kinetic term and no spatial derivatives. Promoting it to a field on space cannot manufacture inertia; the faithful promotion of a first-order reaction law with spatial transport is a *reaction–diffusion* (parabolic) equation, not a wave (hyperbolic) one.

Proposition 3.1 (No gravitational-collapse channel in the Sophia sector)**.** *Because λ is relaxational, its linear inhomogeneities obey a parabolic equation with a real, non-oscillatory spectrum. There is no Jeans-type instability and no gravitational-collapse route to structure in the Sophia sector. Any structure on the Vessel must arise from a different, bistable order parameter.*

V.F.S. does not form structure the way ΛCDM does. Sophia is relaxational, not inertial, so it can only smooth itself out. The only route to structure is morphological symmetry breaking — the folding field (§4).

3.2 Sophia-density perturbations: reaction–diffusion on the expanding $\mathbb{H}^2$

Write $\lambda(t,x) = \bar{\lambda}(t) + \delta\lambda(t,x)$ and linearise about the homogeneous trajectory. With a phenomenological Sophia transport coefficient $D \geq 0$, and noting that physical gradients on the comoving surface redshift by Ω_P^{-2},

$$\partial_t \delta\lambda = F_\lambda(t)\,\delta\lambda + \frac{D}{\Omega_P^2}\,\Delta_h\,\delta\lambda, \qquad F_\lambda = \frac{\partial\dot{\lambda}}{\partial\lambda} = -\Gamma_\lambda, \quad \Gamma_\lambda \geq 0, \tag{3.3}$$

where $\Gamma_\lambda = \gamma + \eta_{\text{fold}} Q_{\text{fold}}$ is the local Sophia relaxation rate (the same rate whose folding-enhancement $\Gamma_\lambda = \gamma + \eta_{\text{fold}} Q_{\text{fold}}$ was found in the stability analysis). The Laplace–Beltrami operator on $\mathbb{H}^2$ has purely continuous spectrum with a gap,

$$\boxed{-\Delta_h\, Y_k = \left(\tfrac{1}{4} + k^2\right) Y_k, \qquad k \in [0,\infty),} \tag{3.4}$$

the celebrated spectral gap $\frac{1}{4}$ of the hyperbolic plane (no mode has eigenvalue below $\frac{1}{4}$). Each mode then evolves by

$$\dot{\delta\lambda}_k = s_k(t)\,\delta\lambda_k, \qquad \boxed{s_k(t) = -\Gamma_\lambda(t) - \frac{D\left(\frac{1}{4} + k^2\right)}{\Omega_P(t)^2}.} \tag{3.5}$$

Proposition 3.2 (Linear homogenisation)**.** *For $\Gamma_\lambda \geq 0$ and $D \geq 0$ every mode has $s_k < 0$: all Sophia inhomogeneities decay. The Vessel homogenises at the linear level. Moreover the hyperbolic gap supplies a minimum diffusive damping $D \cdot \frac{1}{4}/\Omega_P^2 > 0$ even for the smoothest mode $k = 0$: negative spatial curvature actively resists the largest-scale inhomogeneity.*

Integrating,

$$\delta\lambda_k(t) = \delta\lambda_k(0)\,\exp\left[-\int_0^t \Gamma_\lambda\,dt' - D\left(\tfrac{1}{4} + k^2\right)\mathcal{I}(t)\right], \qquad \mathcal{I}(t) := \int_0^t \frac{dt'}{\Omega_P(t')^2}. \tag{3.6}$$

Proposition 3.3 (Finite diffusive smoothing budget)**.** *Under sustained Open-Gate inflow $\lambda \to \lambda_\infty > 0$, the late-time scale factor grows linearly, $\Omega_P \sim \alpha\lambda_\infty t$, so the diffusion integral converges:*

$$\boxed{\mathcal{I}_\infty = \int_0^\infty \frac{dt}{\Omega_P^2} < \infty.} \tag{3.7}$$

Hence diffusion can suppress a mode by at most the finite factor $\exp[-D(\frac{1}{4} + k^2)\mathcal{I}_\infty]$*: the expanding Vessel has a finite total smoothing budget, a comoving analogue of a damping (Silk-like) scale.*

This yields two asymptotic fates, set entirely by the reaction rate Γ_λ:

1. **Generic ($\gamma > 0$).** The reaction integral $\int \Gamma_\lambda\,dt \to \infty$ dominates after diffusion saturates; *all* modes decay to zero. Complete homogenisation.
2. **Margin-stalled ($\Gamma_\lambda \to 0$).** On the active margin where cleansing stalls ($m_* \to 0$, $\inf C_1 = 0$ in the Lyapunov layer), the reaction integral is finite and the modes *freeze* at a scale-dependent residual,
$$\frac{\delta\lambda_k(\infty)}{\delta\lambda_k(0)} = \exp\left[-D\left(\tfrac{1}{4} + k^2\right)\mathcal{I}_\infty\right], \tag{3.8}$$
small scales (high k) erased, large scales (down to the $\frac{1}{4}$ floor) preserved — a frozen inhomogeneity spectrum with a characteristic scale $k_{1/2} \sim \mathcal{I}_\infty^{-1/2}$.

Corollary 3.1 (The smooth Vessel)**.** *Linearly, Sophia structure cannot grow; the Vessel either homogenises completely (sustained cleansing) or freezes a smooth, large-scale residual (stalled margin). There is no spontaneous Sophia clumping.*

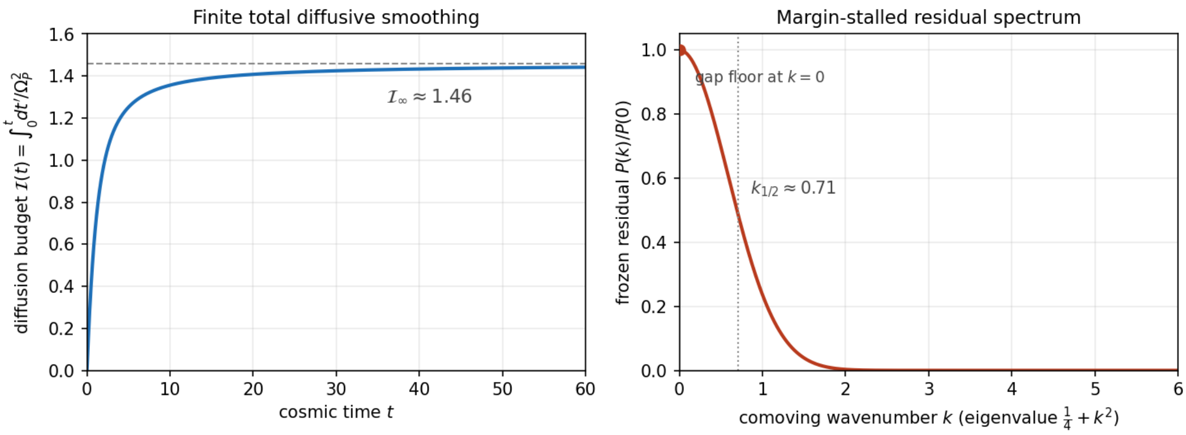

Figure 3.1: Left: the diffusive smoothing budget $\mathcal{I}(t) = \int_0^t dt'/\Omega_P^2$ saturates to a finite $\mathcal{I}_\infty$ (illustrative background $\lambda_0 = 0.4 \to \lambda_\infty = 1$, $\Omega_P(0) = 1$). Right: the margin-stalled frozen residual spectrum $P(k)/P(0)$ (with $D = 0.5$): the hyperbolic gap leaves a $k = 0$ floor while small scales are erased, defining a structure scale $k_{1/2} \approx 0.71$. Parameters illustrative.

3.3 Folding as the order parameter: pattern formation on the Vessel

By Proposition 3.1 structure must come from the bistable folding field. Promote $q_{\text{fold}} \to q_{\text{fold}}(t, x)$ with its Landau gradient dynamics plus transport:

$$\boxed{\partial_t q_{\text{fold}} = A_{\text{fold}}\, q_{\text{fold}} - B_{\text{fold}}\, q_{\text{fold}}^3 + \frac{D_q}{\Omega_P^2}\, \Delta_h\, q_{\text{fold}}.} \tag{3.9}$$

This is a real Ginzburg–Landau / Allen–Cahn equation on the expanding hyperbolic surface, invariant under the morphological reflection $q_{\text{fold}} \to -q_{\text{fold}}$ ($\mathbb{Z}_2$ symmetry, the two orientations of the fold).

Proposition 3.4 (Symmetry-breaking transition)**.** *The homogeneous state $q_{\text{fold}} = 0$ is stable for $A_{\text{fold}} < 0$. At $A_{\text{fold}} = 0$ it undergoes the supercritical pitchfork (the folding bifurcation), and for $A_{\text{fold}} > 0$ two homogeneous phases $q_{\text{fold}} = \pm\sqrt{A_{\text{fold}}/B_{\text{fold}}}$ become stable. A field initialised near $q_{\text{fold}} = 0$ over an extended Vessel then partitions into domains of $+q_*$ and $-q_*$ separated by walls on which $q_{\text{fold}} = 0$.*

These domain walls are the structure: loci of *unfolded* (old) morphology embedded in a folded Vessel — topological defects of the broken $\mathbb{Z}_2$.

Proposition 3.5 (Wall width and tension)**.** *A static planar wall has profile $q_{\text{fold}} = q_* \tanh\!\big(\xi/(\sqrt{2}\,\ell)\big)$ in physical proper distance ξ, with*

$$\boxed{\ell_{\text{phys}} = \sqrt{\frac{D_q}{A_{\text{fold}}}}\quad (\Omega_P\text{-independent}), \qquad \tau \sim \sqrt{D_q}\,\frac{A_{\text{fold}}^{3/2}}{B_{\text{fold}}}.} \tag{3.10}$$

The physical thickness is set by the diffusion-to-reaction ratio and does not stretch with expansion; deeper folding (larger A_{fold}) makes thinner, higher-tension walls.

Sketch. In comoving coordinates the static balance is $\frac{D_q}{\Omega_P^2}\partial_x^2 q_{\text{fold}} = -A_{\text{fold}} q_{\text{fold}} + B_{\text{fold}} q_{\text{fold}}^3$, the standard quartic kink with comoving width $\ell_{\text{com}} = \Omega_P^{-1}\sqrt{D_q/A_{\text{fold}}}$; the proper width $\ell_{\text{phys}} = \Omega_P\,\ell_{\text{com}} = \sqrt{D_q/A_{\text{fold}}}$. The tension is the kink action $\int(\frac{1}{2}D_q|\partial q_{\text{fold}}|^2 + \mathcal{E}_{\text{fold}})$, giving the quoted scaling. □

Corollary 3.2 (Coarsening of the fold network)**.** *Allen–Cahn walls move by mean curvature, $v_n = -M\,\kappa_{\text{curv}}$ with mobility $M \sim D_q/\Omega_P^2$, so the domain network coarsens; simultaneously Hubble flow stretches comoving separations. Structure on the Vessel is therefore a* coarsening domain-wall network *whose characteristic scale grows by both curvature-driven merging and expansion, modified by the underlying $K_h = -1$ geometry (wall geodesics diverge on $\mathbb{H}^2$, accelerating coarsening relative to the flat case).*

The picture is sharp: the folding bifurcation is a phase transition that shatters the Vessel into oriented folded domains; the unfolded walls are the structure; they then coarsen on the expanding hyperbolic surface. This is morphological structure, born of symmetry breaking, not of gravitational clumping.

3.4 Cosmological back-reaction of the fold network

A wall network feeds back on the homogeneous budget. In $2+1$ dimensions a wall is a one-dimensional line in two-dimensional space; a scaling network with one wall per Hubble scale has length $\mathcal{L}_{\rm wall} \sim \Omega_P$ over area $\sim \Omega_P^2$, so

$$\rho_{\rm wall} \sim \frac{\tau\,\mathcal{L}_{\rm wall}}{\Omega_P^2} \propto \Omega_P^{-1}. \tag{3.11}$$

Matching to the $2+1$ dilution law $\rho \propto a^{-2(1+w)}$ gives

$$\boxed{w_{\rm wall} = -\tfrac{1}{2}.} \tag{3.12}$$

The fold network thus contributes a mildly negative-pressure component, intermediate between dust ($w = 0$) and the cosmological-constant/de Sitter limit ($w = -1$). It adds to the reconstructed $w_{\rm eff}(t)$ a folding-structure term that is distinct from the smooth quintessence of the homogeneous Theosis branch.

Corollary 3.3 (Link to the energy-condition question)**.** *Because the network carries $w_{\rm wall} = -\frac{1}{2} > -1$, it is NEC- and WEC-respecting: folded structure is ordinary, non-exotic content. This is the inhomogeneous face of "form makes grace lawful" — where a homogeneous Sophia surge can drive w below -1 (phantom-like, NEC-violating), the embodiment of that excess into a wall network contributes ordinary $w = -\frac{1}{2}$ matter.*

3.5 Open sub-problems and scope

1. **Metric back-reaction.** Lift from test-field to coupled perturbations: let $\delta\lambda$ and $q_{\rm fold}$ source the reconstructed stress tensor and solve the $2+1$ scalar perturbation equations. Expected to preserve Proposition 3.1 (relaxational $\Rightarrow$ no instability) but to dress the wall tension with a geometric self-energy.
2. **Coarsening law.** Derive the domain scale $L(t)$ explicitly on expanding $\mathbb{H}^2$: solve the competition between curvature-driven merging ($L^2 \sim \int M\,dt$) and Hubble stretching, with the hyperbolic correction of Corollary 3.2.
3. **Resurrectio and the network.** A reset rescales Ω_P and shifts $A_{\rm fold}$; it can quench ($A^+_{\rm fold} < 0$) or trigger ($A^+_{\rm fold} > 0$) folding globally, i.e. dissolve or nucleate the entire wall network at a death. The hybrid network history is piecewise-coarsening with reset-induced nucleation events.
4. **Transport constants.** D and D_q are new inputs; a principled derivation (e.g. from a spatial coupling already implicit in the synergy $u = \sqrt{VF + \varepsilon}$) would remove the phenomenological freedom.

Scope. Linear, test-field, single-mode folding, fixed background. The conclusions are: (i) Sophia structure cannot grow (relaxational, homogenising); (ii) structure is morphological — a $\mathbb{Z}_2$ domain-wall network from the folding transition; (iii) the network back-reacts as $w = -\frac{1}{2}$ content. All results inherit the domain-conditional, non-global character of the underlying stability theory.

V.F.S. v2.0 (Open-Gate) · Structure Formation on the Vessel. Perturbation theory on the $2+1$ open FLRW background of the cosmological layer, with the folding field of the geometric layer as order parameter. Propositions are closed-form; corollaries interpretive. Transport constants and single-mode truncation are modelling inputs, not consequences of the base system.

Chapter 4

Spinodal Nucleation of the Fold Network

Scope and epistemic status

A triggering Resurrectio ($A^-_{\text{fold}} < 0 \to A^+_{\text{fold}} > 0$) drives the folding field through its symmetry-breaking transition and nucleates a fresh domain-wall network. This note computes the *initial* density of that network — how many walls per area are born, and at what scale — using the hyperbolic density of states. The two genuinely hyperbolic ingredients are the spectral gap $\frac{1}{4}$ of the Laplacian on $\mathbb{H}^2$ and the Plancherel measure $d\mu(k) = k \tanh(\pi k)\, dk$ that weights the available modes. As before, *Propositions* are closed-form; *Corollaries* interpretive; the single-mode Allen–Cahn truncation and D_q are modelling inputs. Two technical ingredients are verified numerically: the Plancherel moment limit and the Longuet–Higgins nodal coefficient (ratio measured/predicted $= 1.009$ on a Gaussian field).

The field obeys, in comoving coordinates on the unit hyperbolic surface ($K_h = -1$, comoving curvature radius $R_c = 1$),

$$\partial_t q_{\text{fold}} = A_{\text{fold}}\, q_{\text{fold}} - B_{\text{fold}}\, q^3_{\text{fold}} + \frac{D_q}{\Omega_P^2}\, \Delta_h\, q_{\text{fold}}, \qquad -\Delta_h Y_k = \left(\tfrac{1}{4} + k^2\right) Y_k. \tag{4.1}$$

4.1 Linear instability and the nucleation threshold

Immediately after the sudden trigger, A_{fold} jumps to $A^+_{\text{fold}} > 0$ and the homogeneous $q_{\text{fold}} = 0$ left by the previous arc is unstable. Linearising and decomposing in $\mathbb{H}^2$ eigenmodes,

$$\dot{\delta q}_k = \omega_k\, \delta q_k, \qquad \boxed{\omega_k = A^+_{\text{fold}} - \frac{D_q\left(\frac{1}{4} + k^2\right)}{\Omega_P^2}.} \tag{4.2}$$

Introduce the dimensionless trigger strength

$$\boxed{\Lambda := \frac{A^+_{\text{fold}}\, \Omega_P^2}{D_q}.} \tag{4.3}$$

The unstable band is $k < k_{\max}$ with $k^2_{\max} = \Lambda - \frac{1}{4}$ (in units where $D_q/\Omega_P^2 = 1$).

Proposition 4.1 (Nucleation threshold, distinct from the folding threshold)**.** *A spatial mode grows only if $\Lambda > \frac{1}{4}$. Hence there are two thresholds:*

- *folding threshold $A^+_{\text{fold}} > 0$ ($\Lambda > 0$): the homogeneous $q_{\text{fold}} = 0$ is unstable;*
- *nucleation threshold $A^+_{\text{fold}} > D_q/(4\Omega_P^2)$ ($\Lambda > \frac{1}{4}$): some spatial mode is unstable, so domains form.*

In the window $0 < \Lambda < \frac{1}{4}$ the Vessel folds, but uniformly: a single domain, no walls. Only above $\Lambda = \frac{1}{4}$ does a wall network nucleate.

The hyperbolic gap creates a nucleation threshold absent in flat space. A weak trigger ($0 < \Lambda < \frac{1}{4}$) folds the Vessel as a whole, with no internal structure; structure requires $\Lambda > \frac{1}{4}$.

4.2 The amplified field and its spectral moment

Starting from small initial fluctuations δq_0, each mode is amplified by $e^{\omega_k t}$, so the post-quench power spectrum is Gaussian in k,

$$S_k(t) \propto e^{2\omega_k t} \propto e^{-\beta\, k^2}, \qquad \beta = \frac{2D_q t}{\Omega_P^2} = \frac{2t}{\Omega_P^2/D_q}, \tag{4.4}$$

weighted by the hyperbolic Plancherel density of states $d\mu(k) = k \tanh(\pi k)\, dk$ (which reduces to the flat $k\, dk$ for $k \gg 1$ and is suppressed $\sim \frac{\pi}{2} k^2\, dk$ for $k \ll 1$). The relevant spectral second moment is

$$\left\langle \tfrac{1}{4} + k^2 \right\rangle = \tfrac{1}{4} + \frac{\displaystyle\int_0^\infty k^2\, e^{-\beta k^2}\, k \tanh(\pi k)\, dk}{\displaystyle\int_0^\infty e^{-\beta k^2}\, k \tanh(\pi k)\, dk}. \tag{4.5}$$

Proposition 4.2 (Hyperbolic small-scale moment)**.** *In the long-wavelength (hyperbolic-dominated) regime $\beta \gg 1$, where $\tanh(\pi k) \approx \pi k$,*

$$\langle k^2 \rangle \to \frac{3}{2\beta} = \frac{3\Omega_P^2}{4D_q t}. \tag{4.6}$$

Verified: the full integral with $\tanh(\pi k)$ approaches $3/(2\beta)$ as $\beta \to \infty$ (e.g. 0.0285 vs 0.0300 at $\beta = 50$). The Plancherel k^2-suppression at small k is what replaces the flat $1/\beta$ by $3/(2\beta)$ and disfavours the very largest scales.

The set time — when nonlinearity arrests growth at $q_* = \sqrt{A^+_{\text{fold}}/B^+_{\text{fold}}}$ — is the Kibble–Zurek time $t_{\text{set}} = A^{+\,-1}_{\text{fold}} \ln(q_*/\delta q_0)$, giving $\beta_{\text{set}} = 2\Lambda^{-1} L$ with the logarithmic factor $L := \ln(q_*/\delta q_0)$. Hence

$$\boxed{\left\langle \tfrac{1}{4} + k^2 \right\rangle_{\text{set}} = \frac{1}{4} + \frac{3\Lambda}{4L}.} \tag{4.7}$$

4.3 Wall density and the initial domain size

For an isotropic Gaussian field in two dimensions, the expected length of nodal lines (the walls, where $q_{\text{fold}} = 0$) per unit area is the Longuet–Higgins/Berry result

$$\boxed{n_{\text{wall}} = \frac{1}{2\sqrt{2}} \sqrt{\left\langle \tfrac{1}{4} + k^2 \right\rangle},} \tag{4.8}$$

verified numerically here to 0.9%. The initial comoving domain size is $\ell_{\text{init}} \sim 1/n_{\text{wall}}$:

$$\boxed{\ell_{\text{init}} \sim \frac{2\sqrt{2}}{\sqrt{\frac{1}{4} + \frac{3\Lambda}{4L}}} \quad \text{(units of } R_c).} \tag{4.9}$$

Proposition 4.3 (Three regimes of nucleation)**.**

(i) $\Lambda < \frac{1}{4}$*: no nucleation (uniform fold, Proposition 4.1).*

(ii) $\frac{1}{4} < \Lambda \lesssim \Lambda_*$: $\langle \frac{1}{4} + k^2 \rangle \to \frac{1}{4}$ *and* $\ell_{\text{init}} \to 4\sqrt{2}\, R_c$ *— the gap pins the domain size to the curvature radius; only a few curvature-scale domains form. (With* $L \approx 4$*, multi-domain structure* $\ell_{\text{init}} < R_c$ *sets in only at* $\Lambda_* \approx 42$*.)*

(iii) $\Lambda \gg \Lambda_*$: $\ell_{\text{init}} \sim \sqrt{4L/(3\Lambda)}\, R_c$*, i.e. in proper units* $\ell^{\text{phys}}_{\text{init}} \sim \sqrt{(D_q/A^+_{\text{fold}})\, L}$ *— the Kibble–Zurek correlation length* $\xi_+ = \sqrt{D_q/A^+_{\text{fold}}}$ *dressed by the* $\sqrt{L}$ *logarithm.*

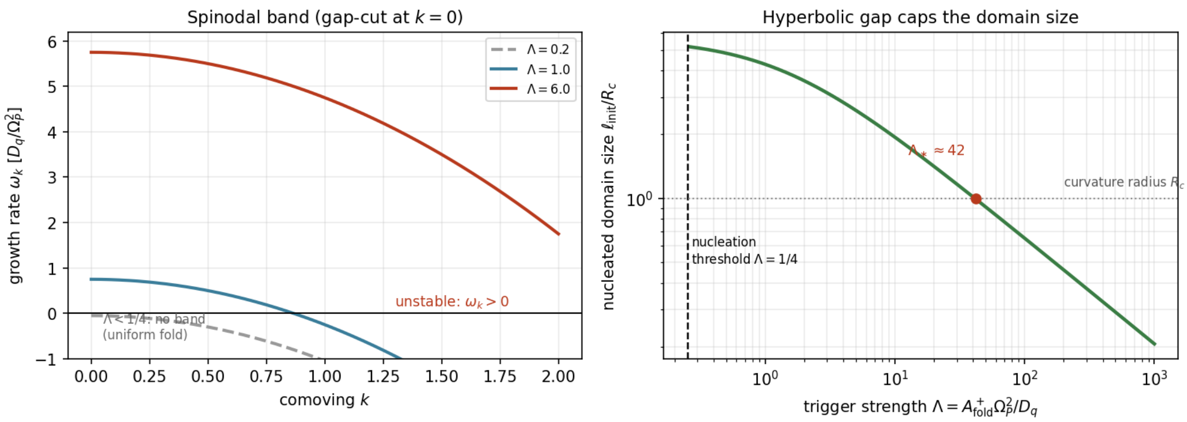

Figure 4.1: Left: linear growth rate ω_k for three trigger strengths. The Laplacian gap shifts the whole curve so that $\omega_0 = \Lambda - \frac{1}{4}$: below $\Lambda = \frac{1}{4}$ ($\Lambda = 0.2$, dashed) there is no unstable band and no nucleation. Right: the nucleated comoving domain size $\ell_{\rm init}/R_c$ versus trigger strength Λ. Near threshold the hyperbolic gap caps the size at the curvature radius (few large domains); genuine multi-domain structure ($\ell_{\rm init} < R_c$) requires $\Lambda > \Lambda_* \approx 42$; for strong triggers $\ell_{\rm init} \sim \sqrt{L/\Lambda}$ recovers the Kibble–Zurek scale. Illustrative $L = \ln(q_*/\delta q_0) = 4$.

4.4 Hyperbolic signatures

Two features distinguish nucleation on $\mathbb{H}^2$ from the flat case, both traceable to the gap and the measure:

Corollary 4.1 (Gap as a nucleation gate and a maximum domain size)**.** *The spectral gap $\frac{1}{4}$ does double duty. As an infrared cutoff it forbids weak-trigger nucleation ($\Lambda < \frac{1}{4}$: uniform fold) and caps the largest nucleated domain at the curvature radius R_c. In flat space any $A^+_{\rm fold} > 0$ nucleates and arbitrarily large domains are allowed at threshold; negative curvature both gates and bounds the structure.*

Corollary 4.2 (Measure suppression of large scales)**.** *Even within the unstable band, the Plancherel weight $k\tanh(\pi k)$ suppresses long-wavelength modes ($\sim \frac{\pi}{2}k^2$ versus the flat k), so large domains are doubly disfavoured: by the gap cutoff and by the density of states. The nucleated network is therefore finer, and more curvature-pinned, than a flat-space estimate would give.*

Interpretive: a gentle resurrection ($\Lambda < \frac{1}{4}$) re-forms the Vessel as a single undivided folded whole; only a sufficiently sharp resurrection ($\Lambda > \Lambda_*$) shatters it into many oriented domains. The negative curvature of the live domain resists fine differentiation — structure is born coarse unless the trigger is strong.

4.5 Placement and open items

This nucleation scale $\ell_{\rm init}$ is the initial condition for the intra-arc coarsening law (subproblem 2): each triggering arc begins at $\ell_{\rm init}$ (set here) and coarsens toward the frozen scale $\sqrt{\ell_0^2 + 2cD_q\mathcal{I}_\infty}$. Consistency check: the strong-trigger regime (iii) reproduces the correlation length $\xi_+ = \sqrt{D_q/A^+_{\rm fold}}$ assumed in subproblem 3, now with the Kibble–Zurek $\sqrt{L}$ dressing and the hyperbolic curvature cap.

Open items.

1. A 2D Allen–Cahn quench simulation on a hyperbolic patch to confirm the nucleated wall count $\propto \langle \frac{1}{4} + k^2 \rangle \propto \Lambda$ and fix the $O(1)$ coefficients.
2. Defect statistics beyond the mean: the distribution of initial domain areas (the Gauss–Bonnet collapse law of subproblem 2 then selects which survive).
3. The exact crossover near $\Lambda = \frac{1}{4}$ where the gap and the band compete — a hyperbolic finite-size/zero-mode analysis.

Scope. Linear spinodal stage, Gaussian-field defect counting, single-mode Allen–Cahn, sudden quench; D_q and the noise level δq_0 are modelling inputs. Results: a hyperbolic nucleation threshold $\Lambda > \frac{1}{4}$ distinct from the folding threshold; a closed-form initial wall density $n_{\mathrm{wall}} = \frac{1}{2\sqrt{2}}\sqrt{\frac{1}{4} + 3\Lambda/4L}$; and a curvature-capped domain size interpolating from R_c (near threshold) to the Kibble–Zurek $\sqrt{(D_q/A^{+}_{\mathrm{fold}})L}$ (strong trigger). The Plancherel moment and the nodal coefficient are numerically verified. Domain-conditional throughout.

V.F.S. v2.0 (Open-Gate) · Spinodal Nucleation of the Fold Network on $\mathbb{H}^2$. Linear instability of the Allen–Cahn folding field after a triggering reset, with the hyperbolic spectral gap and Plancherel measure. Propositions closed-form (Plancherel moment and Longuet–Higgins coefficient numerically verified); corollaries interpretive; transport constant and noise level are modelling inputs.

Chapter 5

Intra-Arc Coarsening of the Fold Network

Scope and epistemic status

The companion notes established that structure on the Vessel is a $\mathbb{Z}_2$ domain-wall network of the folding field and that a Resurrectio reset acts on it in three regimes, giving a sawtooth network-scale history. This note supplies the missing intra-arc piece: *between* deaths, how does the network scale $L_{\text{phys}}(t)$ evolve under pure Allen–Cahn coarsening on the expanding hyperbolic surface? No resets are involved here. As before, *Propositions* are closed-form, *Corollaries* interpretive; the single-mode Allen–Cahn truncation and the transport constant D_q are modelling inputs.

The field obeys, in comoving coordinates on the unit hyperbolic surface ($K_h = -1$),

$$\partial_t q_{\text{fold}} = A_{\text{fold}}\, q_{\text{fold}} - B_{\text{fold}}\, q_{\text{fold}}^3 + \frac{D_q}{\Omega_P^2}\, \Delta_h\, q_{\text{fold}}. \tag{5.1}$$

The folding parameters $A_{\text{fold}}, B_{\text{fold}}$ are taken quasi-stationary on the coarsening timescale (deep in a folded arc, $A_{\text{fold}} > 0$), so the dynamics is genuine coarsening of established domains rather than the initial transition.

Remark (Dimensionality matters). Allen–Cahn coarsening is curvature-driven only in two or more spatial dimensions, giving the $\ell \sim t^{1/2}$ law; in one dimension kink–antikink attraction is exponentially weak and coarsening is merely logarithmic. The 1D space–time proxy used to illustrate the reset regimes is therefore *not* representative of the true 2D coarsening rate. The present law is genuinely two-dimensional, derived via the geodesic-curvature flow on $\mathbb{H}^2$.

5.1 Wall mobility on the expanding background

In the sharp-interface limit of (5.1), a domain wall moves along its comoving normal with velocity proportional to its geodesic curvature, with a mobility set by the (comoving) diffusion coefficient:

$$v_n^{\text{com}} = -M(t)\,\kappa_g, \qquad M(t) = \frac{D_q}{\Omega_P(t)^2}. \tag{5.2}$$

The scale-factor suppression Ω_P^{-2} is the redshift of comoving gradients: as the Vessel expands, the *comoving* mobility falls, even though the proper wall structure is preserved (the proper wall width $\sqrt{D_q/A_{\text{fold}}}$ is Ω_P-independent). It is convenient to trade time for the accumulated budget already central to the smoothing analysis,

$$\mathcal{I}(t) := \int_0^t \frac{dt'}{\Omega_P(t')^2}, \qquad \frac{d}{dt} = \frac{1}{\Omega_P^2}\frac{d}{d\mathcal{I}}, \tag{5.3}$$

which is finite, $\mathcal{I} \to \mathcal{I}_\infty < \infty$, under sustained inflow ($\Omega_P \sim \alpha\lambda_\infty t$).

5.2 Gauss–Bonnet: the exact single-domain law

For a single simply-connected domain of comoving area $\mathcal{A}$ bounded by a wall undergoing geodesic curve-shortening on $\mathbb{H}^2$, the Gauss–Bonnet theorem gives the total geodesic curvature $\oint \kappa_g\, ds = 2\pi - \int_{\text{enc}} K\, dA = 2\pi + \mathcal{A}$ (since $K_h = -1$). The enclosed area therefore obeys

$$\boxed{\frac{d\mathcal{A}}{dt} = -\frac{D_q}{\Omega_P^2}\big(2\pi + \mathcal{A}\big) \quad \Longleftrightarrow \quad \frac{d\mathcal{A}}{d\mathcal{I}} = -D_q\big(2\pi + \mathcal{A}\big),} \tag{5.4}$$

with the closed-form solution

$$\boxed{\mathcal{A}(\mathcal{I}) = \big(\mathcal{A}_0 + 2\pi\big)e^{-D_q\mathcal{I}} - 2\pi.} \tag{5.5}$$

Proposition 5.1 (Survival threshold and frozen scale)**.** *A domain collapses ($\mathcal{A} = 0$) at budget $\mathcal{I}_* = D_q^{-1}\ln(1 + \mathcal{A}_0/2\pi)$. Because the total budget is finite, a domain survives the entire arc iff $\mathcal{I}_* > \mathcal{I}_\infty$, i.e.*

$$\boxed{\mathcal{A}_0 > \mathcal{A}_{\text{frozen}} := 2\pi\big(e^{D_q\mathcal{I}_\infty} - 1\big).} \tag{5.6}$$

Domains smaller than $\mathcal{A}_{\text{frozen}}$ collapse before the budget runs out; larger ones freeze into the comoving lattice. $\mathcal{A}_{\text{frozen}}$ is the characteristic frozen comoving domain area.

Proposition 5.2 (Hyperbolic acceleration)**.** *For $\mathcal{A} \gg 2\pi$ the law is $\dot{\mathcal{A}} \approx -M\mathcal{A}$: large domains collapse* exponentially*, in contrast to the flat case $\dot{\mathcal{A}} = -2\pi M$ (constant-rate Gage–Hamilton shrinking). Negative spatial curvature thus accelerates the disappearance of large domains, sharpening coarsening relative to flat space.*

In the small-budget regime $D_q\mathcal{I}_\infty \ll 1$, $\mathcal{A}_{\text{frozen}} \approx 2\pi D_q\mathcal{I}_\infty$, so the frozen comoving length is $\ell_{\text{com}} \sim \sqrt{2\pi D_q\mathcal{I}_\infty}$ — matching the scaling law below. The hyperbolic exponential (Proposition 5.2) only bites once domains grow toward the comoving curvature radius ($\mathcal{A} \sim 2\pi$), i.e. when the budget is large enough.

5.3 The network scaling law and its saturation

For the network, the standard Lifshitz–Allen–Cahn scaling $\ell\,\dot{\ell} \sim M$ becomes, on the budget variable,

$$\frac{d}{dt}\ell_{\text{com}}^2 \simeq 2c\,\frac{D_q}{\Omega_P^2} \quad \Longrightarrow \quad \boxed{\ell_{\text{com}}^2(t) = \ell_{\text{com}}^2(0) + 2c\,D_q\,\mathcal{I}(t),} \tag{5.7}$$

with an $O(1)$ constant c (consistent with Proposition 5.1, $c \sim \pi$).

Proposition 5.3 (Comoving coarsening freezes)**.** *Since $\mathcal{I}(t) \to \mathcal{I}_\infty < \infty$, the comoving domain size saturates,*

$$\ell_{\text{com}}(\infty) = \sqrt{\ell_{\text{com}}^2(0) + 2c\,D_q\,\mathcal{I}_\infty} < \infty. \tag{5.8}$$

Coarsening stops in comoving coordinates: the wall network freezes into the comoving lattice after a finite amount of merging. This is the morphological counterpart of the finite diffusive smoothing budget governing Sophia inhomogeneities — both are controlled by the same $\mathcal{I}_\infty$.

The expanding Vessel has a single finite "activity budget" $\mathcal{I}_\infty = \int dt/\Omega_P^2$. It caps both how much Sophia inhomogeneity can diffuse away and how much the fold network can coarsen. After it is spent, comoving structure is frozen.

5.4 The physical structure scale: two regimes

The proper (physical) network scale combines comoving coarsening with expansion,

$$\boxed{L_{\text{phys}}(t) = \Omega_P(t)\,\ell_{\text{com}}(t) = \Omega_P(t)\sqrt{\ell_{\text{com}}^2(0) + 2c\,D_q\,\mathcal{I}(t)}.} \tag{5.9}$$

Proposition 5.4 (Coarsening transient, then Hubble dilution)**.** *There are two regimes, separated by the budget-saturation time t_{sat} (where $\mathcal{I}$ levels):*

(i) *Early ($t \lesssim t_{\text{sat}}$): both factors grow, so L_{phys} rises faster than the scale factor — genuine coarsening on top of stretch.*

(ii) *Late ($t \gtrsim t_{\text{sat}}$): $\ell_{\text{com}} \to$ const, so $L_{\text{phys}} \to \ell_{\text{com}}(\infty)\,\Omega_P(t) \propto \Omega_P \propto t$ — pure Hubble dilution of a frozen comoving pattern.*

Coarsening is an early-time transient with a finite budget; the late-time growth of structure is expansion, not merging.

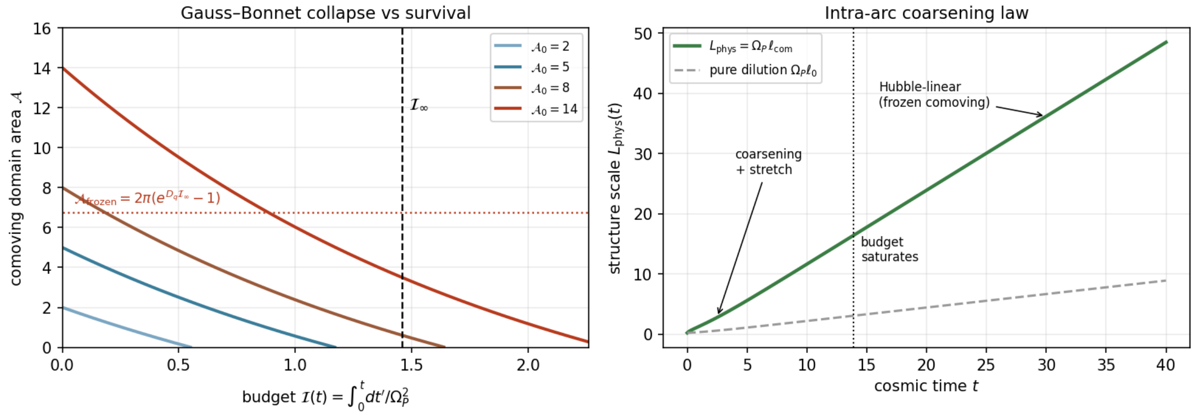

Figure 5.1: Left: Gauss–Bonnet single-domain area $\mathcal{A}(\mathcal{I})$ for several initial sizes (exact closed form, verified). Domains below $\mathcal{A}_{\text{frozen}} = 2\pi(e^{D_q\mathcal{I}_\infty} - 1)$ collapse before the finite budget $\mathcal{I}_\infty$ is spent; larger ones survive and freeze. Right: the physical structure scale $L_{\text{phys}}(t) = \Omega_P \ell_{\text{com}}$ (solid) shows an early coarsening-plus-stretch phase pulling above pure dilution $\Omega_P \ell_0$ (dashed), then a late Hubble-linear regime once the budget saturates. Background and (D_q, c) illustrative; $\mathcal{I}_\infty \approx 1.46$.

5.5 Placement in the sawtooth and open items

This intra-arc curve is the rising tooth of the hybrid sawtooth from subproblem 3: within each life-arc $L_{\text{phys}}(t)$ follows Proposition 5.4 (coarsen, then dilate), and at each death the reset acts discretely — diluting upward, refining to $\sqrt{D_q/A_{\text{fold}}}$, or erasing. Composing the intra-arc law with the reset map gives the long-run fixed-point scale, the natural next step.

Open items.

1. Determine the $O(1)$ constant c and confirm the $t^{1/2}$ (budget) law by a 2D Allen–Cahn field simulation on a hyperbolic patch with a clean structure-factor scale estimator.
2. Compose this intra-arc law with the reset map to obtain the sawtooth fixed-point structure scale, using the non-Zeno dwell time as the arc length.
3. Track the $w = -\frac{1}{2}$ wall-network contribution to the reconstructed $w_{\text{eff}}(t)$ over an arc, since $\rho_{\text{wall}} \propto \tau/\Omega_P$ with τ quasi-stationary and Ω_P growing.

Scope. Single life-arc, no resets, quasi-stationary folding parameters, single-mode Allen–Cahn; D_q a modelling input. Results: an exact Gauss–Bonnet single-domain law with a survival threshold and hyperbolic acceleration; a network scaling law $\ell_{\text{com}}^2 = \ell_0^2 + 2cD_q\mathcal{I}$ that freezes with the finite budget $\mathcal{I}_\infty$; and a physical scale L_{phys} that coarsens early and dilutes linearly late. Domain-conditional throughout.

V.F.S. v2.0 (Open-Gate) · Intra-Arc Coarsening of the Fold-Domain Network. The single-domain Gauss–Bonnet law is exact (closed form, numerically verified); the network scaling law is standard Allen–Cahn on the budget variable. Transport constant and single-mode truncation are modelling inputs; the constant c awaits a 2D field measurement.

Chapter 6

Resurrectio and the Fold-Domain Network

Scope and epistemic status

The companion note showed that structure on the Vessel is morphological: a $\mathbb{Z}_2$ domain-wall network of the folding field $q_{\text{fold}}(t,x)$, obeying the Allen–Cahn equation on the expanding hyperbolic surface. This note asks how the hybrid Resurrectio reset $\mathfrak{R}$ acts on that network. The key observation is that $\mathfrak{R}$ is *homogeneous* — it acts on the global state $(\sigma, \lambda, \Omega_P)$ — and, as established in the Lyapunov layer, it does *not* act on q_{fold}. So a reset shifts the folding potential uniformly across the whole Vessel while leaving the field configuration $q_{\text{fold}}(x)$ pointwise intact. Whether structure survives, dilutes, or is born depends on the post-reset sign of the activation index. As before, *Propositions* are closed-form; *Corollaries* interpretive; the single-mode Allen–Cahn truncation and the transport constant D_q are modelling inputs.

The network field obeys

$$\partial_t q_{\text{fold}} = A_{\text{fold}}\, q_{\text{fold}} - B_{\text{fold}}\, q_{\text{fold}}^3 + \frac{D_q}{\Omega_P^2}\,\Delta_h\, q_{\text{fold}}, \qquad q_* = \sqrt{Q^*_{\text{fold}}}, \quad Q^*_{\text{fold}} = A_{\text{fold}}/B_{\text{fold}}, \tag{6.1}$$

and the reset (geometric convention, with $m_R = (1-q_R)\sigma^-$ the metabolised resistance) is

$$\mathfrak{R}: \quad \sigma^+ = \sigma^- - m_R, \quad \lambda^+ = \lambda^- + m_R, \quad \Omega_P^+ = \Omega_P^- + \kappa_R \mathcal{G}^-_{\text{recepta}}, \quad q_{\text{fold}} \text{ continuous}, \tag{6.2}$$

which shifts the activation index by $\Delta A_{\text{fold}} = \Theta\, m_R + c_1 \Delta I_{\text{gate}}$, $\Theta = c_0 - c_1(\gamma + k_\sigma)$, exactly as in the folding-trigger note.

6.1 Three regimes of a reset acting on the network

Because $\mathfrak{R}$ shifts A_{fold} uniformly while preserving $q_{\text{fold}}(x)$, the network's fate is fixed by the pre- and post-reset signs of A_{fold}:

Definition 6.1 (Reset regimes)**.**

(I) **Quench** ($A^-_{\text{fold}} > 0 \to A^+_{\text{fold}} < 0$): the double well collapses to a single well at $q_{\text{fold}} = 0$.

(II) **Rescale** ($A^-_{\text{fold}} > 0 \to A^+_{\text{fold}} > 0$): the wells persist with shifted depth and the scale factor jumps.

(III) **Trigger** ($A^-_{\text{fold}} < 0 \to A^+_{\text{fold}} > 0$): a single well bifurcates into a double well.

Proposition 6.1 (Regime boundaries)**.** *With $\Delta A_{\text{fold}} = \Theta m_R + c_1 \Delta I_{\text{gate}}$, a reset from A^-_{fold} produces $A^+_{\text{fold}} = A^-_{\text{fold}} + \Delta A_{\text{fold}}$. Hence, neglecting the bounded gate term: a quench requires $\Theta < 0$ and $|\Theta| m_R > A^-_{\text{fold}}$; a trigger requires $\Theta > 0$ and $\Theta m_R > |A^-_{\text{fold}}|$; otherwise the reset rescales. The folding-response coefficient Θ thus selects whether resurrection destroys, preserves, or creates morphological structure.*

A death-and-resurrection does not move the walls. It lifts or lowers the entire folding potential at once — and depending on the sign of Θ, the whole network dissolves, is re-tensioned and diluted, or is nucleated from nothing.

6.2 Quench: dissolution of the network

When $A^{+}_{\text{fold}} < 0$, the configuration $q_{\text{fold}}(x)$ everywhere relaxes to $q_{\text{fold}} = 0$. Linearising (6.1) about zero, each mode decays at rate $|A^{+}_{\text{fold}}| + D_q(\frac{1}{4} + k^2)/\Omega_P^{+2} > 0$, so the entire pattern fades on the timescale

$$\tau_{\text{dissolve}} = \frac{1}{|A^{+}_{\text{fold}}|}. \tag{6.3}$$

The walls do not migrate and annihilate; the whole differentiated morphology evaporates uniformly, returning the Vessel to the undifferentiated old form.

Corollary 6.1 (Death as leveling)**.** *A quench-resurrection returns the entire Vessel to the unfolded state $q_{\text{fold}} = 0$: all oriented domains are erased, all walls dissolved. Death levels morphological structure.*

6.3 Trigger: nucleation as a sudden quench (Kibble–Zurek)

When $A^{-}_{\text{fold}} < 0 \to A^{+}_{\text{fold}} > 0$, the homogeneous $q_{\text{fold}} = 0$ left behind by the previous arc becomes unstable. Since $\mathfrak{R}$ is instantaneous, A_{fold} jumps discontinuously: a *sudden quench* through the symmetry- breaking transition. Linearising about $q_{\text{fold}} = 0$ after the jump,

$$\dot{\delta q_k} = \left[A^{+}_{\text{fold}} - \frac{D_q(\frac{1}{4} + k^2)}{\Omega_P^{+2}}\right] \delta q_k, \tag{6.4}$$

so a band of modes $k < k_{\text{max}}$, $k^2_{\text{max}} = A^{+}_{\text{fold}}\Omega_P^{+2}/D_q - \frac{1}{4}$, grows and the field undergoes spinodal decomposition. The emergent (physical) domain size is the post-quench correlation length,

$$\boxed{\ell_{\text{init}} \sim \xi_+ = \sqrt{\frac{D_q}{A^{+}_{\text{fold}}}}} \tag{6.5}$$

the same scale as the wall width: the network is born saturated — as fine as the walls themselves — then coarsens. Deeper triggers (larger A^{+}_{fold}) nucleate finer networks.

Corollary 6.2 (Resurrection as individuation)**.** *A trigger-resurrection shatters the undifferentiated old form into a fine network of newly differentiated oriented domains, with the old form surviving only as the thin walls between them. Where quench levels, trigger differentiates.*

This is a Kibble–Zurek-type defect formation: the reset drives the system through a symmetry-breaking transition and freezes in a defect network whose initial scale is set by the correlation length at the transition. Here the quench is sudden, so ℓ_{init} is the spinodal scale ξ_+.

6.4 Rescale: re-tensioning and dilution

When both $A^{\pm}_{\text{fold}} > 0$, the domains persist but everything jumps. With $q_* = \sqrt{A_{\text{fold}}/B_{\text{fold}}}$, $\ell_{\text{phys}} = \sqrt{D_q/A_{\text{fold}}}$, $\tau \sim \sqrt{D_q}\, A^{3/2}_{\text{fold}}/B_{\text{fold}}$, a reset induces

$$q^{+}_{*} = \sqrt{\frac{A^{+}_{\text{fold}}}{B^{+}_{\text{fold}}}}, \qquad \frac{\ell^{+}_{\text{phys}}}{\ell^{-}_{\text{phys}}} = \sqrt{\frac{A^{-}_{\text{fold}}}{A^{+}_{\text{fold}}}}, \qquad \frac{\tau^{+}}{\tau^{-}} = \left(\frac{A^{+}_{\text{fold}}}{A^{-}_{\text{fold}}}\right)^{3/2} \frac{B^{-}_{\text{fold}}}{B^{+}_{\text{fold}}}, \tag{6.6}$$

and, crucially, the comoving topology is untouched while proper inter-wall separations are stretched by the scale-factor jump,

$$\boxed{L^{+}_{\text{phys}} = \frac{\Omega_P^{+}}{\Omega_P^{-}} L^{-}_{\text{phys}} = \left(1 + \frac{\kappa_R \mathcal{G}^{-}_{\text{recepta}}}{\Omega_P^{-}}\right) L^{-}_{\text{phys}}.} \tag{6.7}$$

The network is diluted: the same domains, spread wider.

Corollary 6.3 (Amplitude convergence under repeated rescaling)**.** *The contraction $Q^{*+}_{\text{fold}} - Q_{\text{crit}} = \kappa_{\text{fold}}(Q^{*-}_{\text{fold}} - Q_{\text{crit}})$ of the folding-trigger note drives the domain amplitude to a universal value across repeated resets, $q_* \to \sqrt{Q_{\text{crit}}}$ (when $\sum m_R = \infty$). Resurrection stabilises* how strongly *the Vessel is folded, while expansion keeps diluting* how widely*.*

6.5 The hybrid network history

Over a sequence of life-arcs the structure scale $L_{\text{phys}}(t)$ follows a sawtooth: within each arc the network coarsens (curvature-driven merging plus Hubble stretch), and at each death it is acted on by $\mathfrak{R}$ — diluted upward (rescale), reset to the fine scale ξ_+ (trigger), or erased (quench).

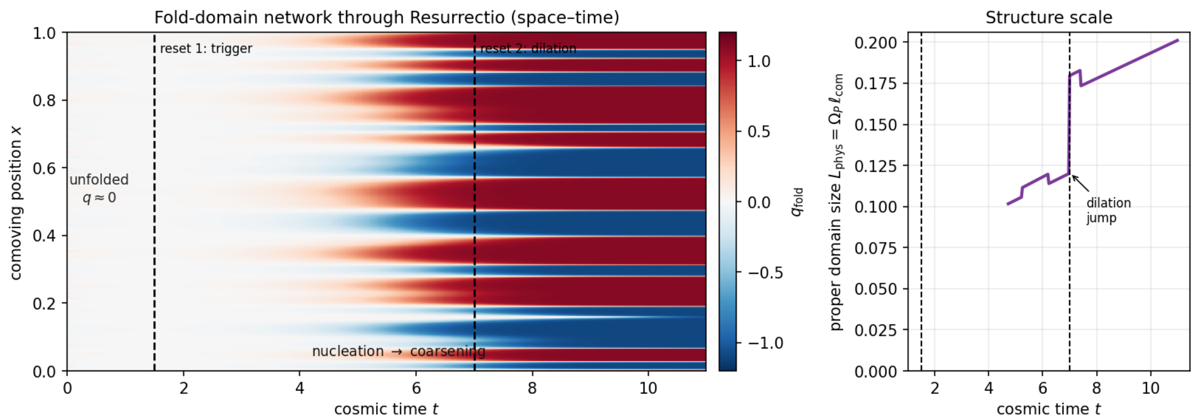

Figure 6.1: Allen–Cahn simulation of the fold-domain network through two resets (illustrative parameters; 1D comoving slice as a proxy). Left: space–time map of q_{fold}. The Vessel is unfolded ($q_{\text{fold}} \approx 0$) until *reset 1* triggers the transition; domains then nucleate from fluctuations and coarsen. *Reset 2* is a rescale (regime II): the comoving pattern is untouched but the scale factor jumps. Right: the proper domain size $L_{\text{phys}} = \Omega_P\, \ell_{\text{com}}$ grows by coarsening and shows a discrete dilation jump at reset 2.

Corollary 6.4 (Resurrection as a renormalisation step on structure)**.** *Each reset is a coarse-grain (dilation, regime II) or refine (nucleation to ξ_+, regime III) operation on the network, composed with intra-arc coarsening. The long-run structure scale is the fixed point of (arc coarsening) $\circ$ (reset action); a steady balance between expansion-driven dilution and reset-driven nucleation.*

6.6 Cosmological back-reaction across resets

Within a scaling arc the wall network contributes $\rho_{\text{wall}} \sim \tau/\Omega_P \propto \Omega_P^{-1}$, i.e. $w_{\text{wall}} = -\frac{1}{2}$ in $2{+}1$ dimensions (companion note). At a reset this jumps: τ changes through $A_{\text{fold}}, B_{\text{fold}}$ and Ω_P jumps, so

$$\frac{\rho^+_{\text{wall}}}{\rho^-_{\text{wall}}} = \frac{\tau^+}{\tau^-} \cdot \frac{\Omega^-_P}{\Omega^+_P}. \tag{6.8}$$

A quench sets $\rho_{\text{wall}} \to 0$: the morphological energy is released back into the homogeneous Sophia budget. A trigger creates ρ_{wall} from zero: structure energy is drawn from the folding transition. Thus resurrection redistributes energy between the homogeneous (λ) and structured (wall-network) sectors — a discrete exchange invisible to the homogeneous reconstruction alone.

6.7 Reading and open items

Interpretive reading. The walls are loci of persistent unfolded (old) form. A quench- resurrection returns the whole Vessel to undifferentiated old form (death as leveling, $\Theta < 0$); a trigger-resurrection shatters that undifferentiated form into newly differentiated oriented domains, the old form surviving only as the thin boundaries between them (resurrection as individuation, $\Theta > 0$). Repeated rescaling fixes the depth of folding ($q_* \to \sqrt{Q_{\text{crit}}}$) while expansion keeps widening the pattern. The sign of Θ — whether metabolised resistance promotes or suppresses form — decides which face a given resurrection wears.

Open items.

1. Quantify the spinodal nucleation density on $\mathbb{H}^2$ (hyperbolic spectral gap modifies the unstable band) and the resulting initial wall count per Hubble area.
2. Derive the sawtooth fixed-point scale explicitly from (coarsening law) $\circ$ (reset map), using the non-Zeno dwell time as the arc length.
3. Track the $\lambda \leftrightarrow$ network energy exchange as a correction to the reconstructed $w_{\text{eff}}(t)$ history, and test for a folding-induced feature near the phantom divide.

Scope. Single-mode Allen–Cahn, test-field, 1D simulation proxy; transport constant D_q a modelling input. Conclusions: a homogeneous reset acts on the morphological network in three regimes fixed by sign Θ and the jump size — dissolve, dilute, or nucleate — and resurrection functions as a renormalisation step on Vessel structure, exchanging energy between the homogeneous and structured sectors. Domain-conditional throughout.

V.F.S. v2.0 (Open-Gate) · Resurrectio and the Fold-Domain Network. Builds on the structure-formation companion (Allen–Cahn folding network on $2{+}1$ *open FLRW) and the Resurrectio reset of the Lyapunov layer. Propositions closed-form; corollaries interpretive; numerical figure illustrative.*

Chapter 7

The Sawtooth Fixed Point of the Fold Network

Scope and epistemic status

The three preceding notes supply the pieces: the nucleation scale ℓ_{init} at a triggering reset (subproblem 1), the intra-arc coarsening law governed by the finite budget $\mathcal{I}_\infty = \int_0^\infty dt/\Omega_P^2$ (subproblem 2), and the three-regime reset map — quench, rescale, trigger (subproblem 3). This note composes them: over a sequence of life-arcs punctuated by resets, what is the long-run network scale? The central result is that the finite coarsening budget forces the *comoving* scale to converge to a constant ℓ_* regardless of the (random) reset schedule, so the physical scale is asymptotically $L_{\text{phys}}(t) \to \ell_* \Omega_P(t) \propto t$. This also resolves the earlier worry that the random arc length would obstruct a fixed point: the finite budget makes the asymptotics universal. As before, *Propositions* are closed-form; *Corollaries* interpretive; the reduced recursion (tracking the scale, not the full field), the coarsening constant c, and the reset-regime schedule are modelling inputs.

7.1 The comoving recursion

Track the comoving network scale ℓ_{com} across arcs. Index the arcs by j; let ℓ_j be the comoving scale entering arc j and $\Delta\mathcal{I}_j = \int_{\text{arc j}} dt/\Omega_P^2$ the budget consumed during it. Within the arc, the coarsening law (subproblem 2) gives $\ell^2 \mapsto \ell^2 + 2cD_q\,\Delta\mathcal{I}_j$; at the closing reset, the regime acts:

$$\boxed{\ell_{j+1}^2 = R_j\Big(\ell_j^2 + 2cD_q\,\Delta\mathcal{I}_j\Big), \qquad R_j = \begin{cases} \text{id}, & \text{rescale (comoving topology unchanged)} \\ \ell_{\text{init}}^2, & \text{trigger (re-nucleation)} \\ \varnothing, & \text{quench (dissolve; re-nucleate at next trigger)} \end{cases}} \tag{7.1}$$

The essential constraint, inherited from the finite expansion budget, is

$$\sum_j \Delta\mathcal{I}_j \le \mathcal{I}_\infty < \infty. \tag{7.2}$$

Remark (Why rescale is the identity on the comoving scale). A rescale reset jumps Ω_P but leaves the comoving pattern intact, so it does not change ℓ_{com}; its only effect is on the physical scale, $L_{\text{phys}} = \Omega_P \ell_{\text{com}}$. The comoving recursion therefore sees rescales as no-ops, triggers as resets to ℓ_{init}, and quenches as dissolutions.

7.2 Finite budget forces convergence

Proposition 7.1 (Vanishing late-time coarsening)**.** *For any reset schedule with non-Zeno dwell $\Delta t_j \ge \Delta t_* > 0$, the per-arc coarsening increment vanishes,*

$$2cD_q\,\Delta\mathcal{I}_j \to 0 \qquad (j \to \infty), \tag{7.3}$$

because $\sum_j \Delta\mathcal{I}_j \le \mathcal{I}_\infty < \infty$ *forces* $\Delta\mathcal{I}_j \to 0$ *(equivalently* $\Delta\mathcal{I}_j \le \Delta t_j/\Omega_P^2$ *with* $\Omega_P \to \infty$*). The random arc lengths do not affect this: the finite budget dominates.*

Proposition 7.2 (Convergence of the comoving scale)**.** *The comoving scale converges to a constant* ℓ_**:*

(i) if triggers recur infinitely often, each pins ℓ *to* ℓ_{init} *and (by Proposition 7.1) the subsequent coarsening cannot grow it, so* $\ell_j \to \ell_{\text{init}}$*;*

(ii) if all resets after some arc J *are rescales, then* $\ell_j^2 \to \ell_J^2 + 2cD_q \sum_{j \ge J} \Delta\mathcal{I}_j =: \ell_\infty^2 < \infty$*, the frozen value.*

In either case $\ell_j \to \ell_* \in [\ell_{\text{init}}, \ell_\infty]$*, with* $\ell_\infty = \sqrt{\ell_{\text{init}}^2 + 2cD_q(\mathcal{I}_\infty - \mathcal{I}_{\text{nuc}})}$ *the maximal frozen scale (single nucleation, no retrigger).*

Proposition 7.3 (Universal physical asymptotics)**.** *The physical structure scale obeys*

$$\boxed{L_{\text{phys}}(t) = \Omega_P(t)\,\ell_{\text{com}}(t) \;\longrightarrow\; \ell_*\,\Omega_P(t) \;\propto\; t,} \tag{7.4}$$

independent of the reset schedule. Different random histories differ only in the value of $\ell_* \in [\ell_{\text{init}}, \ell_\infty]$*, not in the* $\propto \Omega_P$ *growth.*

> The coarsening budget is finite, so the comoving network freezes. Resurrection can *refresh* the scale (a trigger resets it to ℓ_{init}) but cannot *re-coarsen* it once the budget is spent. The late-time structure scale is a relic of the young Vessel, carried outward by pure expansion.

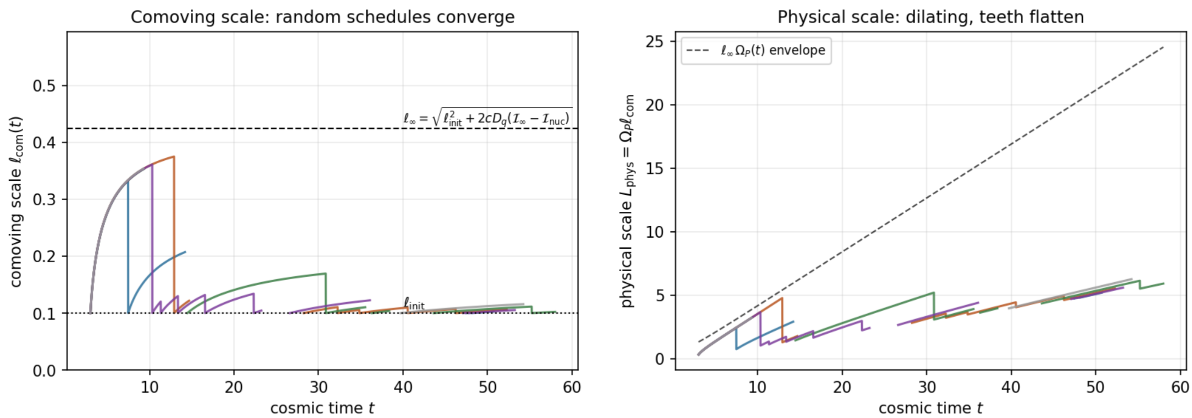

Figure 7.1: Reduced recursion over five random reset schedules (non-Zeno spacing; rescale/trigger/quench with probabilities $0.7/0.15/0.15$; illustrative $2cD_q = 0.5$, $\ell_{\text{init}} = 0.1$, $\mathcal{I}_\infty \approx 1.46$). Left: the comoving scale $\ell_{\text{com}}(t)$ — early arcs coarsen substantially toward ℓ_∞, but as the budget depletes the teeth flatten and every schedule is pinned near ℓ_{init} at late times. Right: the physical scale $L_{\text{phys}} = \Omega_P \ell_{\text{com}}$ dilates along the $\ell_\infty \Omega_P(t)$ envelope; gaps are quench intervals (network dissolved). Comoving teeth flatten; physical scale keeps dilating $\propto \Omega_P \propto t$.

7.3 Reading and consequences

Corollary 7.1 (Structure scale is a young-Vessel relic)**.** *Because coarsening is budget-limited, the comoving structure scale is set in the early, slowly-expanding era and frozen thereafter. The mature Vessel does not generate new structure scale by merging; it merely carries the early-frozen pattern outward by expansion. Any apparent late-time growth of* L_{phys} *is Hubble dilution, not coarsening.*

Corollary 7.2 (Resurrection refreshes but cannot re-coarsen)**.** *A late trigger-resurrection re-nucleates the network at the fine scale ℓ_{init}, and the depleted budget leaves it there: the renewed structure is permanently fine. Only resurrections occurring early, while the budget is ample, can be followed by appreciable coarsening. Early death-and-rebirth permits maturation of form; late death-and-rebirth yields perpetually fine, uncoarsened structure.*

The composition closes the programme's internal loop: subproblem 1 sets the floor ℓ_{init}, subproblem 2 sets the ceiling ℓ_∞ via the budget, subproblem 3 supplies the reset action, and the finite budget collapses the sawtooth onto $\ell_* \in [\ell_{\text{init}}, \ell_\infty]$ with universal physical asymptotics $L_{\text{phys}} \propto \Omega_P$.

7.4 Open items and scope

Open items.

1. Replace the reduced scale-recursion by a full 2D Allen–Cahn field simulation through a reset sequence, to confirm ℓ_* and the flattening of the comoving teeth.
2. Couple the reset-regime probabilities to the actual $A_{\text{fold}}(t)$ history (rising with λ via epektasis, dipped by metabolism when $\Theta < 0$), replacing the illustrative fixed probabilities by the dynamically determined regime sequence.
3. Feed ℓ_* and the wall tension into the $w = -\frac{1}{2}$ network back-reaction to obtain the folding-structure contribution to $w_{\text{eff}}(t)$ over cosmic history (link to the energy-condition direction).

Scope. Reduced recursion (network scale, not the full field); single-mode Allen–Cahn; coarsening constant c and reset-regime schedule are modelling inputs; the finite-budget convergence is robust to the random schedule. Result: the comoving network scale converges to $\ell_* \in [\ell_{\text{init}}, \ell_\infty]$ and the physical scale is universally $L_{\text{phys}} \to \ell_* \Omega_P \propto t$. Domain-conditional throughout.

V.F.S. v2.0 (Open-Gate) · The Sawtooth Fixed Point of the Fold Network. Composes the nucleation scale (subproblem 1), the budget-limited coarsening law (subproblem 2), and the three-regime reset map (subproblem 3). The convergence follows from the finite budget $\mathcal{I}_\infty$ and is robust to the random reset schedule. Propositions closed-form; corollaries interpretive; reduced recursion and coarsening constant are modelling inputs.

Chapter 8

The Transport Constants from First Principles

Scope and epistemic status

The structure-formation programme introduced two transport coefficients — a Sophia diffusivity D and a folding diffusivity D_q — as phenomenological inputs, on which all quantitative results depend. This note asks whether VFS mechanics fixes them. The answers are sharp: (i) the Sophia field has *no* fundamental diffusion, $D = 0$, because it is purely relaxational; its inhomogeneities are slaved to the folding field through the embodiment feedback. (ii) The folding diffusivity is *not* an independent constant: its spatial structure (Ω_P^{-2} and the hyperbolic $K_h = -1$) is inherited from the live-domain metric, and its single scalar magnitude is the bending stiffness of the shape-stability operator $\mathcal{L}_0$ — the gradient partner of the rigidity a_0 already in the theory. As before, *Propositions* are closed-form; *Corollaries* interpretive. The slaving relation is verified symbolically; the reduction of D_q to a single bending length still rests on the minimal one-modulus model of $\mathcal{L}_0$, which is flagged.

8.1 Sophia does not diffuse

In the base dynamical system the Sophia production law

$$\dot{\lambda} = (\delta u - \gamma)\tanh(\kappa\sigma) + I_{\text{gate}} - \eta_{\text{fold}}\, q_{\text{fold}}^2 \lambda \tag{8.1}$$

contains no spatial derivative of λ: it is a pointwise relaxation law. Promoting λ to a field therefore introduces no gradient coupling of its own.

Proposition 8.1 (Zero fundamental Sophia diffusivity)**.** *The Sophia field carries no kinetic or gradient term in the base system, so its fundamental diffusivity vanishes, $D = 0$. Linear Sophia inhomogeneities obey a pure local relaxation, $\partial_t \delta\lambda = -\Gamma_\lambda\, \delta\lambda + ($folding source$)$, with no k-dependent spatial transport.*

Remark (Correction to the structure-formation note). The frozen residual spectrum of the structure-formation companion assumed $D > 0$. With $D = 0$, the Sophia-only modes decay uniformly at the local rate Γ_λ with no k-dependence: on a stalled margin ($\Gamma_\lambda \to 0$) they freeze with a *flat* residual, not a scale-graded one. All scale-dependent structure is morphological, not Sophianic.

Although Sophia does not diffuse, it is not inert: the embodiment feedback $-\eta_{\text{fold}} q_{\text{fold}}^2 \lambda$ couples it to the folding field. Linearising about $(\bar{\lambda}, \bar{q})$,

$$\partial_t \delta\lambda = -\big(\Gamma_\lambda + \eta_{\text{fold}}\bar{q}^2\big)\delta\lambda - 2\eta_{\text{fold}}\bar{\lambda}\bar{q}\, \delta q_{\text{fold}}, \tag{8.2}$$

and in the quasi-static (fast-relaxation) limit

$$\boxed{\delta\lambda^* = -\frac{2\eta_{\text{fold}}\bar{\lambda}\bar{q}}{\Gamma_\lambda + \eta_{\text{fold}}\bar{q}^2}\, \delta q_{\text{fold}}.} \tag{8.3}$$

Proposition 8.2 (Sophia structure is slaved to form)**.** *Sophia inhomogeneity is a fixed multiple of folding inhomogeneity, with a bounded coefficient* $S(\bar{q}) = 2\eta_{\text{fold}}\bar{\lambda}\bar{q}/(\Gamma_\lambda + \eta_{\text{fold}}\bar{q}^2)$ *maximised at* $\bar{q} = \sqrt{\Gamma_\lambda/\eta_{\text{fold}}}$*, where* $S_{\max} = \bar{\lambda}\sqrt{\eta_{\text{fold}}/\Gamma_\lambda}$*. Sophia carries no independent structural degree of freedom: its spatial pattern is the shadow of the folding pattern.*

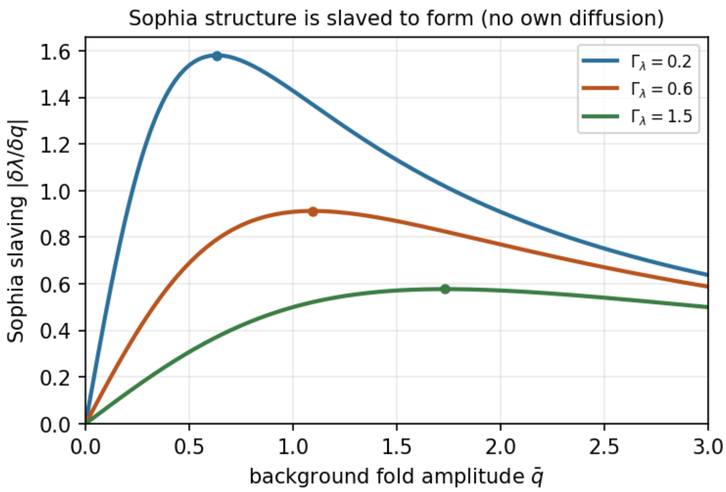

Figure 8.1: The Sophia slaving coefficient $|\delta\lambda/\delta q_{\text{fold}}| = S(\bar{q})$ (verified symbolically). It is bounded, vanishes at $\bar{q} = 0$ and at large $\bar{q}$, and peaks at $\bar{q} = \sqrt{\Gamma_\lambda/\eta_{\text{fold}}}$. Sophia inhomogeneity is sourced entirely by folding inhomogeneity; there is no independent Sophia diffusion to set. Illustrative $\eta_{\text{fold}} = 0.5$, $\bar{\lambda} = 1$.

There is really one structural field. Sophia does not diffuse; wherever the Vessel is inhomogeneous in Sophia, it is because it is inhomogeneous in *form*, and the two are locked by the embodiment feedback. The Sophia diffusivity D is not a missing constant — it is identically zero.

8.2 The folding diffusivity is geometric

The folding field's spatial coupling is fixed in two stages: its *structure* by the metric, its *magnitude* by the shape operator.

8.2.1 Structure: inherited from the live-domain metric

The folding free energy, built geometrically, measures gradients with the live-domain metric g (geometry layer, scalar curvature $-2/\Omega_P^2$, conformally hyperbolic $g = \Omega_P^2 h$):

$$\mathcal{F}[q_{\text{fold}}] = \int_\Sigma \Big[\tfrac{\kappa_b}{2}\,|\nabla_g q_{\text{fold}}|^2 + \tfrac{r}{2} q_{\text{fold}}^2 + \tfrac{B}{4} q_{\text{fold}}^4\Big]\sqrt{g}\,d^2x, \qquad r = a_0 - c_0\lambda - c_1\dot{\lambda} = -A_{\text{fold}}. \tag{8.4}$$

The Laplace–Beltrami operator of g is $\Delta_g = \Omega_P^{-2}\Delta_h$, so the gradient-flow dynamics $\partial_t q_{\text{fold}} = -M\,\delta\mathcal{F}/\delta q_{\text{fold}}$ is automatically

$$\partial_t q_{\text{fold}} = A_{\text{fold}}\,q_{\text{fold}} - B_{\text{fold}}\,q_{\text{fold}}^3 + \frac{M\kappa_b}{\Omega_P^2}\,\Delta_h\,q_{\text{fold}}, \qquad A_{\text{fold}} = -Mr,\; B_{\text{fold}} = MB. \tag{8.5}$$

Proposition 8.3 (The Ω_P^{-2} and $K_h = -1$ are not chosen)**.** *The redshift factor* Ω_P^{-2} *and the hyperbolic Laplacian* Δ_h *in the folding transport term are the Laplace–Beltrami operator of the live-domain metric established in the geometric layer, not modelling choices. Only a single scalar,* $D_q = M\kappa_b$*, remains free.*

8.2.2 Magnitude: the bending stiffness of $\mathcal{L}_0$

The remaining scalar is $D_q = M\kappa_b$, the product of the shape mobility M and the bending stiffness κ_b. The bending stiffness is the gradient coefficient of the very shape-stability operator $\mathcal{L}_0$ whose constant-mode eigenvalue is the rigidity a_0 (the operator whose bifurcation defines folding). In the natural time units where the uniform folding rate is A_{fold} one has $M = 1$, so $D_q = \kappa_b$, and all physical lengths depend only on the ratio

$$\boxed{\ell_b^2 := \frac{\kappa_b}{a_0} = \frac{D_q}{a_0}\,, \qquad \text{wall width} = \sqrt{\frac{D_q}{A_{\text{fold}}}} = \ell_b\sqrt{\frac{a_0}{A_{\text{fold}}}}\,.} \tag{8.6}$$

Proposition 8.4 (Folding diffusivity is the gradient partner of the rigidity)**.** *D_q is not independent of the theory's existing constants: it is the gradient coefficient of $\mathcal{L}_0$, whose constant-mode coefficient is a_0. Physical scales (wall width, nucleation scale, coarsening and frozen scales) depend only on the bending length $\ell_b = \sqrt{D_q/a_0}$, a ratio internal to $\mathcal{L}_0$. In the minimal one-modulus (Helfrich/Willmore) model of $\mathcal{L}_0$, where rigidity and bending share a single modulus, ℓ_b is fixed by the reference shape and $D_q = a_0\ell_b^2$ ceases to be a free input altogether.*

The reduction is concrete: from "an arbitrary spatial coupling plus a free constant" to "a single bending length ℓ_b, the gradient partner of the rigidity a_0 already present in $A_{\text{fold}} = c_0\lambda + c_1\dot{\lambda} - a_0$." The spatial form is pure geometry; the magnitude is the Vessel's stiffness, not a new dial.

8.3 Consequences for the programme

Corollary 8.1 (One structural field, one length)**.** *The structure-formation programme has, after this reduction, a single genuine input: the bending length $\ell_b = \sqrt{D_q/a_0}$. The Sophia diffusivity is zero; the folding transport structure is geometric; the folding magnitude is the shape stiffness. Every scale derived earlier — the nucleation size $\ell_{\text{init}} \sim \ell_b\sqrt{a_0/A^+_{\text{fold}}}\,(\ldots)$, the frozen comoving scale, the sawtooth fixed point — is expressed through ℓ_b and the already-existing folding parameters.*

Corollary 8.2 (Sophia structure rides on form)**.** *Because $\delta\lambda$ is slaved to δq_{fold} (Proposition 8.2), the Sophia inhomogeneity spectrum is the folding spectrum times the bounded factor $S(\bar{q})$. Wherever folding nucleates a domain-wall network, a matching Sophia pattern appears; where folding is quenched, Sophia re-homogenises. There is no separate Sophia structure-formation problem.*

8.4 Open items and scope

Open items.

1. Fix the bending length ℓ_b from an explicit form of the shape-stability operator $\mathcal{L}_0$ (e.g. the second variation of a Willmore/Helfrich bending energy of the live surface), turning Proposition 8.4 from a relation into a number.
2. Verify the slaving relation (Proposition 8.2) beyond the quasi-static limit, including the finite Sophia relaxation time, and check it against a coupled $(q_{\text{fold}}, \lambda)$ field simulation.
3. Propagate $D = 0$ and the slaving into the cosmological back-reaction: the Sophia-structure contribution to w_{eff} is then a fixed multiple of the wall-network contribution.

Scope. Results: the Sophia diffusivity is identically zero (relaxational field), with Sophia inhomogeneity slaved to folding via the embodiment feedback (symbolically verified); the folding transport structure is the Laplace–Beltrami operator of the live-domain metric (not a choice); and the folding magnitude reduces to the bending length $\ell_b = \sqrt{D_q/a_0}$, the gradient partner of the rigidity a_0 in $\mathcal{L}_0$. The remaining modelling freedom is the single length ℓ_b (a number only in the minimal one-modulus model). Domain-conditional throughout.

V.F.S. v2.0 (Open-Gate) · The Transport Constants from First Principles. Closes the structure-formation programme's modelling inputs: $D = 0$ with Sophia slaved to form; D_q geometric in structure and equal to the shape-operator bending stiffness in magnitude, reducible to one bending length. Propositions closed-form (slaving verified symbolically); corollaries interpretive; the one-modulus reduction of $\mathcal{L}_0$ is a flagged model choice.